

JENKINS NOTES

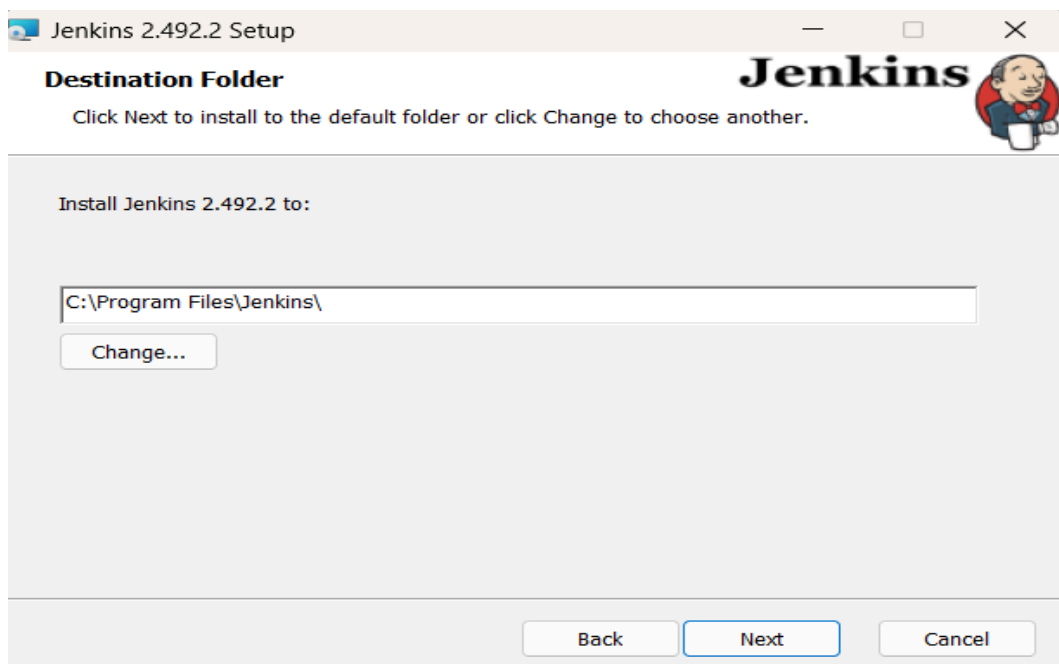
JENKINS INSTALLATION

Step 1 : Download Jenkins from <https://www.jenkins.io/download/> - Windows Users should download the ".msi" file and not ".war" file

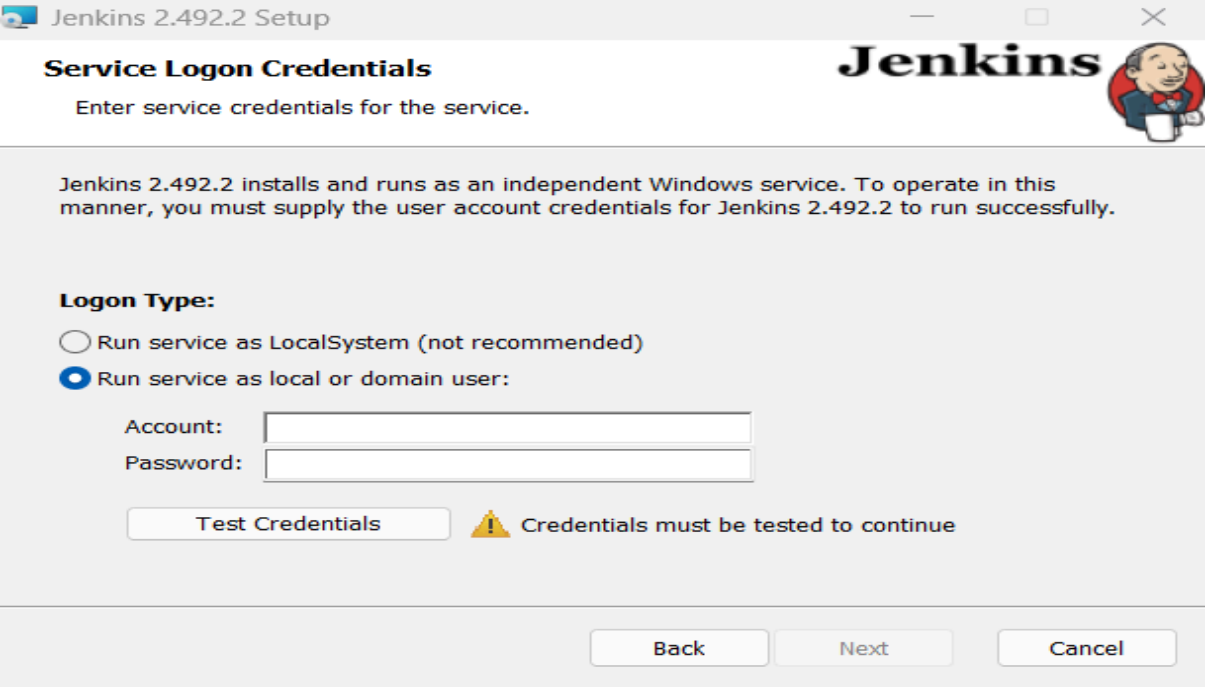
Step 2: Go to download and open Jenkins

+ New ▾ ✂ 📄 📁 🗑️ 🔍 📧 📧 🗑️ ⬆️ Sort ▾ ☰ View ▾ ⋮					
🏠 Home	Name	Date modified	Type	Size ▾	
🖼️ Gallery	▼ Today				
> 📁 OneDrive - Persona	📁 jenkins	16-03-2025 07:47	Windows Installer ...	1,00,424 KB	
	▼ Earlier this week				

Step 3: Click Next ----> Next



Step 4: In "Logon Type" select "Run Services as local system" and click Next



Jenkins 2.492.2 Setup

Service Logon Credentials

Enter service credentials for the service.

Jenkins 2.492.2 installs and runs as an independent Windows service. To operate in this manner, you must supply the user account credentials for Jenkins 2.492.2 to run successfully.

Logon Type:

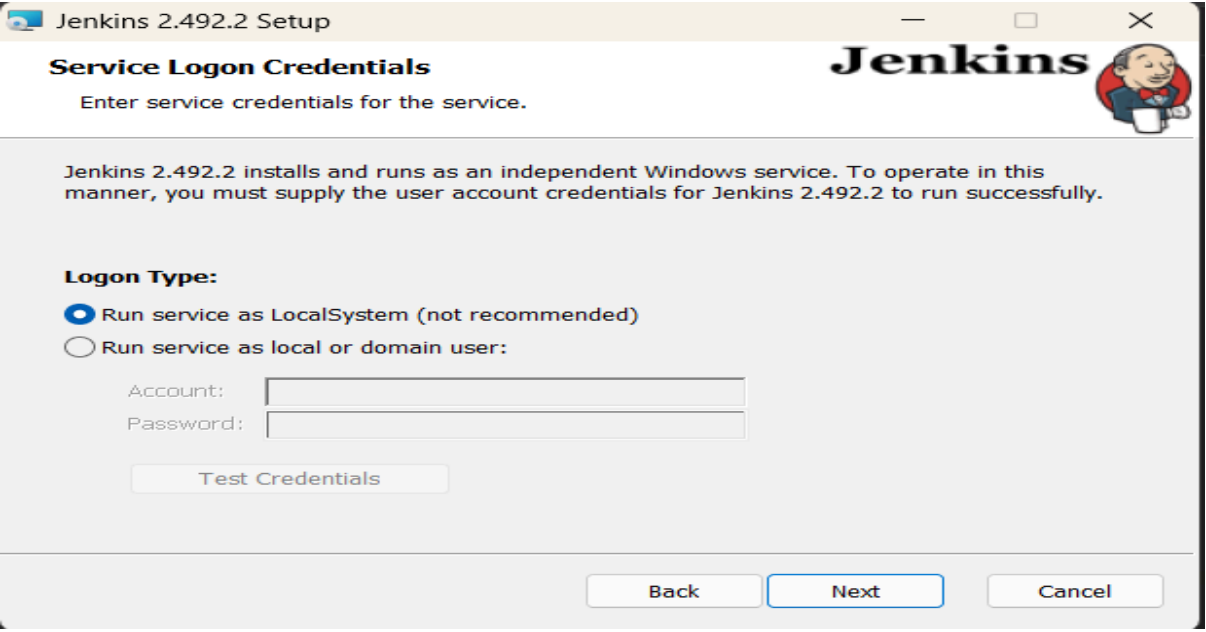
☐ Run service as LocalSystem (not recommended)

☒ Run service as local or domain user:

Account:

Password:

 Credentials must be tested to continue



Jenkins 2.492.2 Setup

Service Logon Credentials

Enter service credentials for the service.

Jenkins 2.492.2 installs and runs as an independent Windows service. To operate in this manner, you must supply the user account credentials for Jenkins 2.492.2 to run successfully.

Logon Type:

☒ Run service as LocalSystem (not recommended)

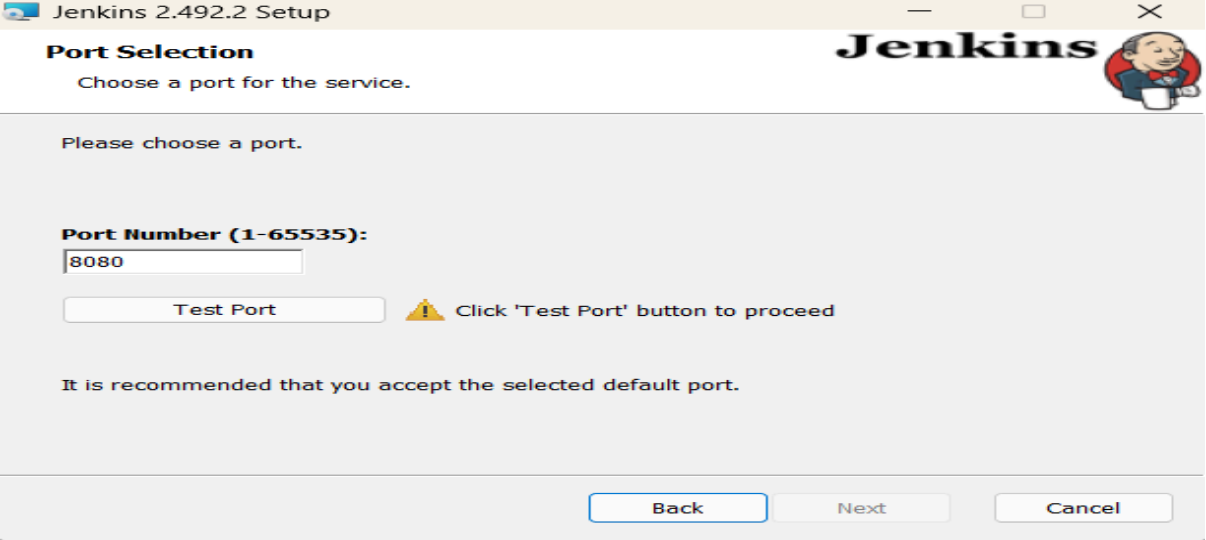
☐ Run service as local or domain user:

Account:

Password:

Step 5: Here we have to test the port , Click on the “Test Port” - If after clicking you get a green colour tick then you port is test is passed , otherwise if not getting the green colour tick then try changing the port and click TEST PORT again and then click next

Note: Remember the port number for future use



Jenkins 2.492.2 Setup

Port Selection

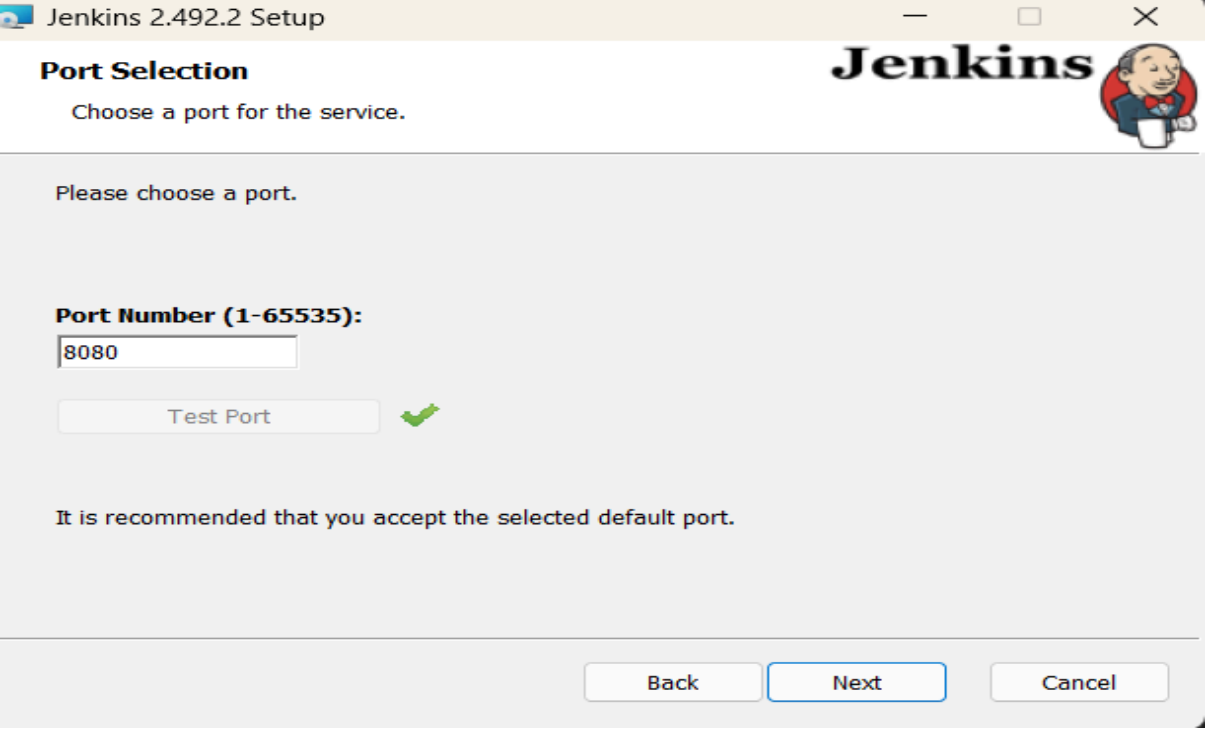
Choose a port for the service.

Please choose a port.

Port Number (1-65535):
8080

 Click 'Test Port' button to proceed

It is recommended that you accept the selected default port.



Jenkins 2.492.2 Setup

Port Selection

Choose a port for the service.

Please choose a port.

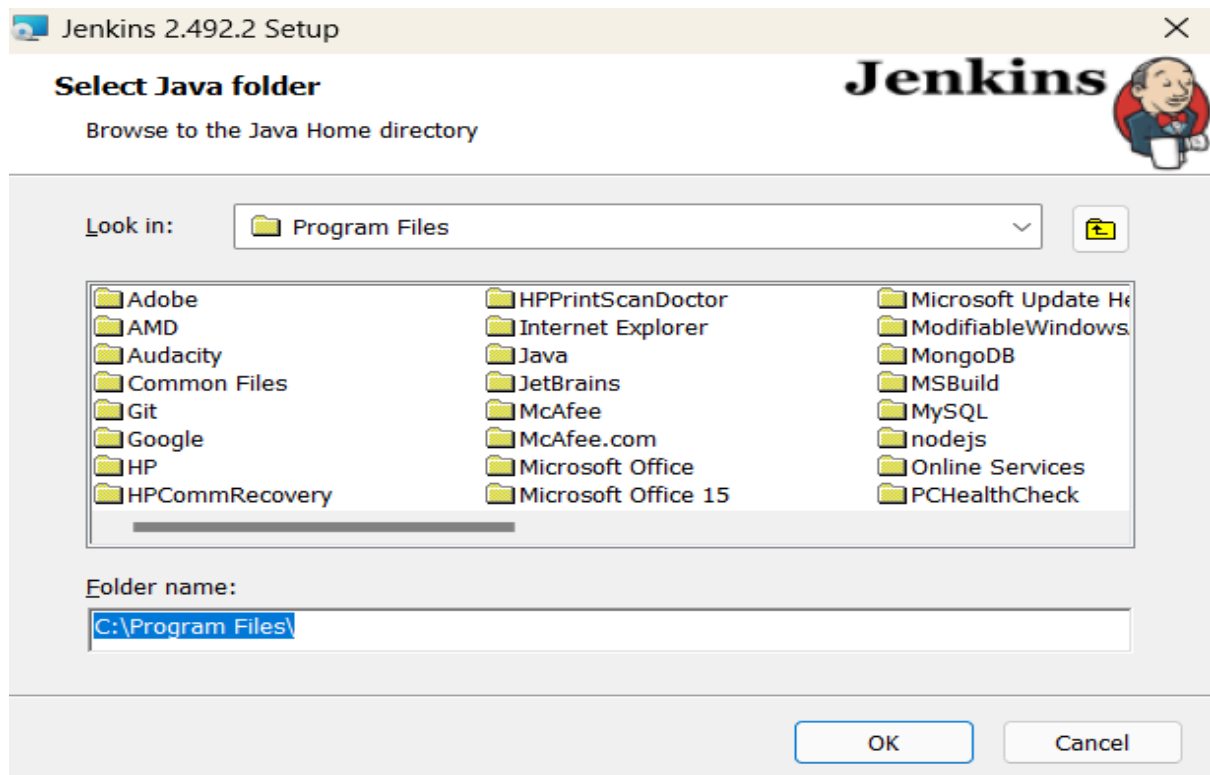
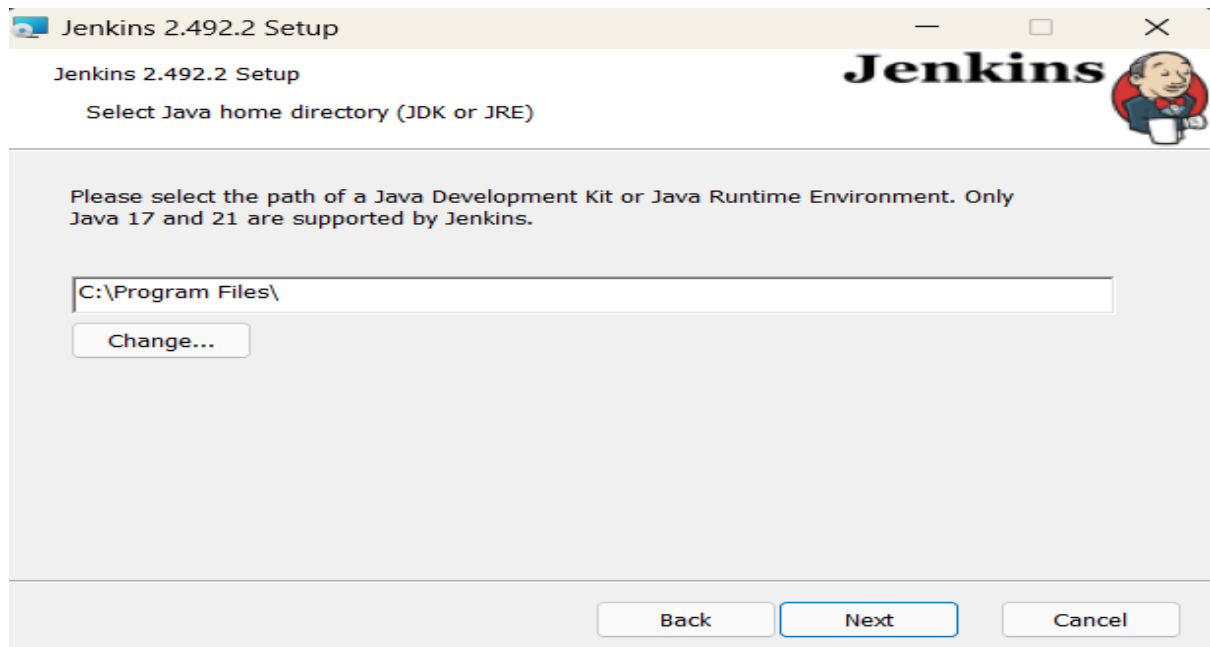
Port Number (1-65535):
8080

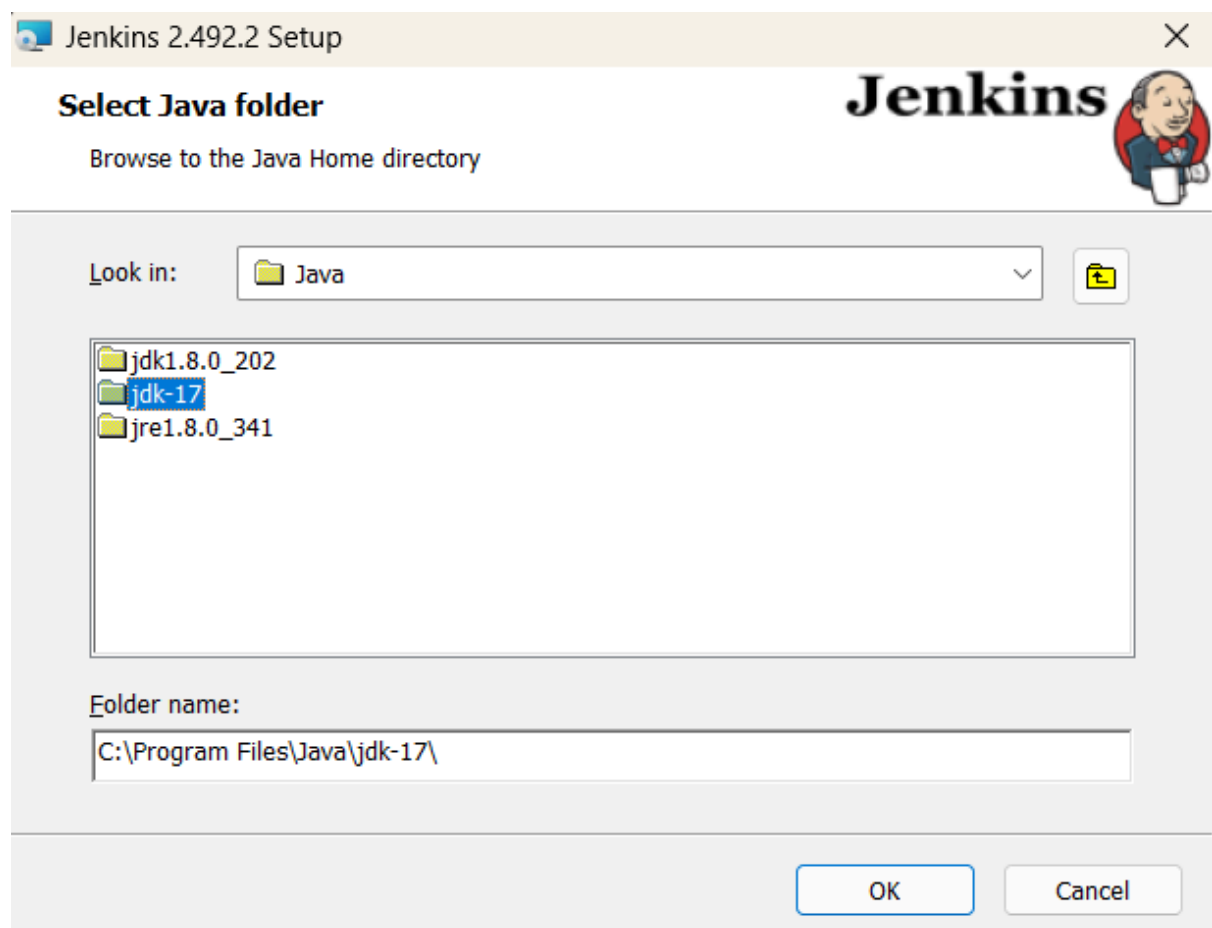
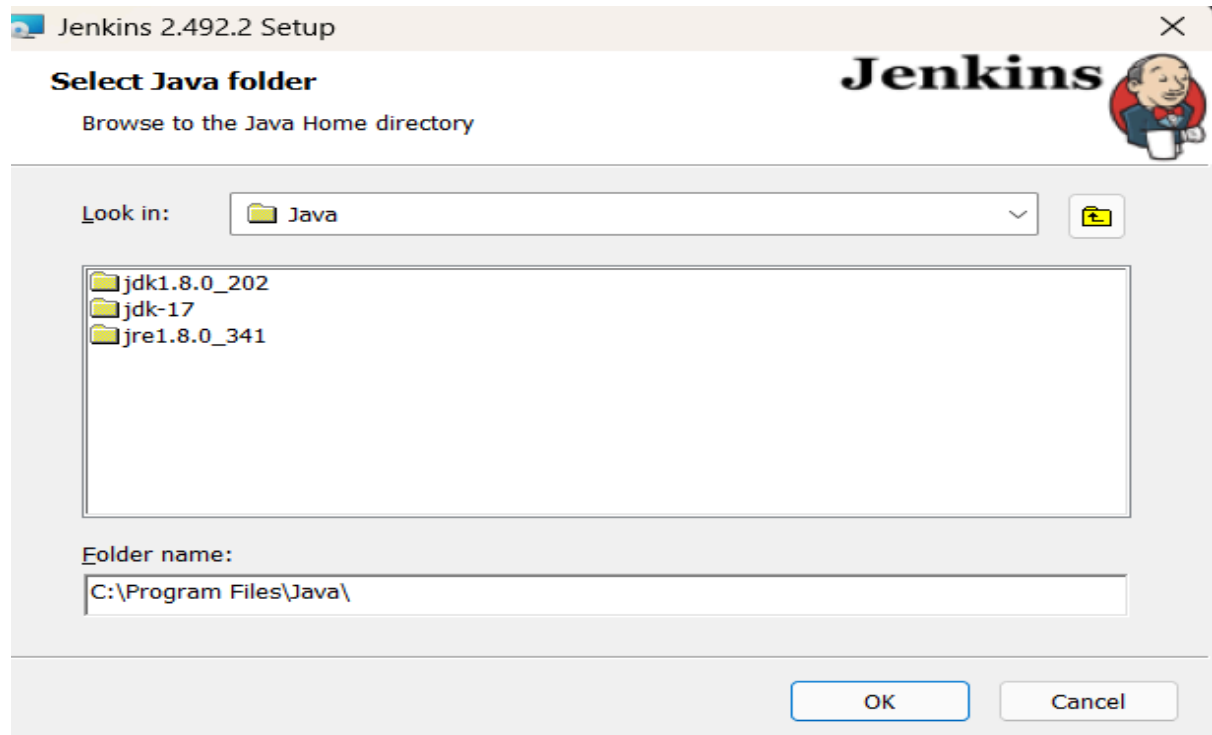


It is recommended that you accept the selected default port.

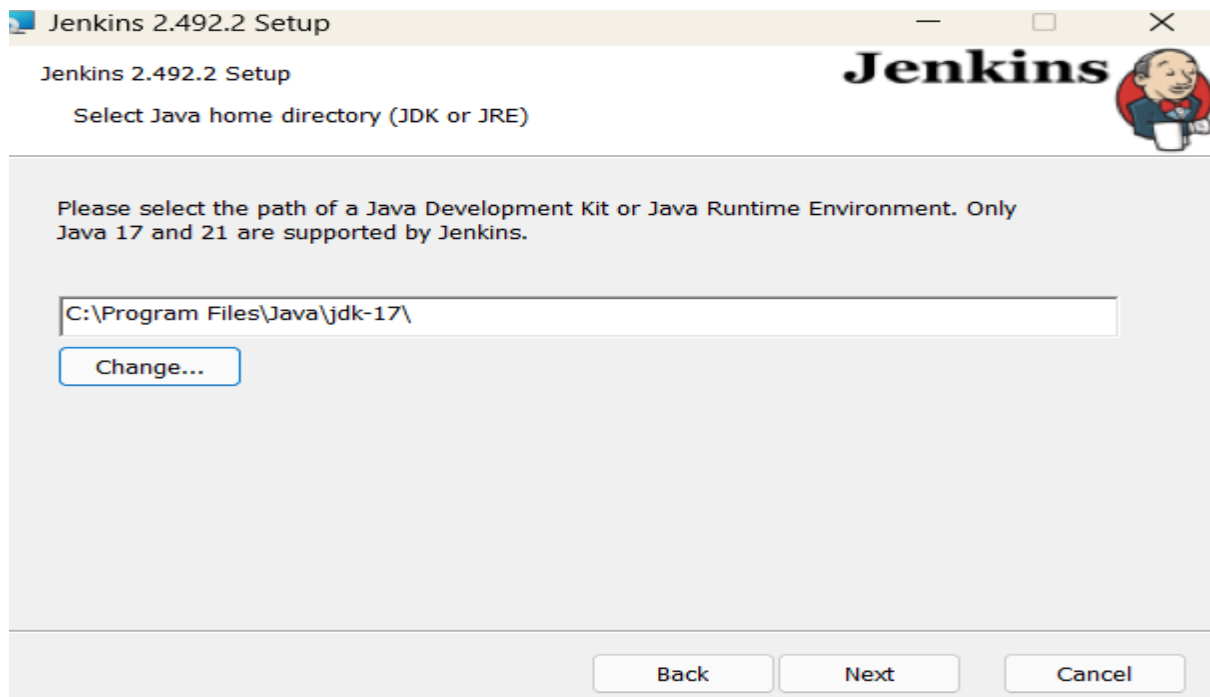
Step 6: Next you have to select the JDK path. Click on Change button ----->select java -----> Select JDK17 -----> click Next

Note: Only JDK 17 and 21 are supported

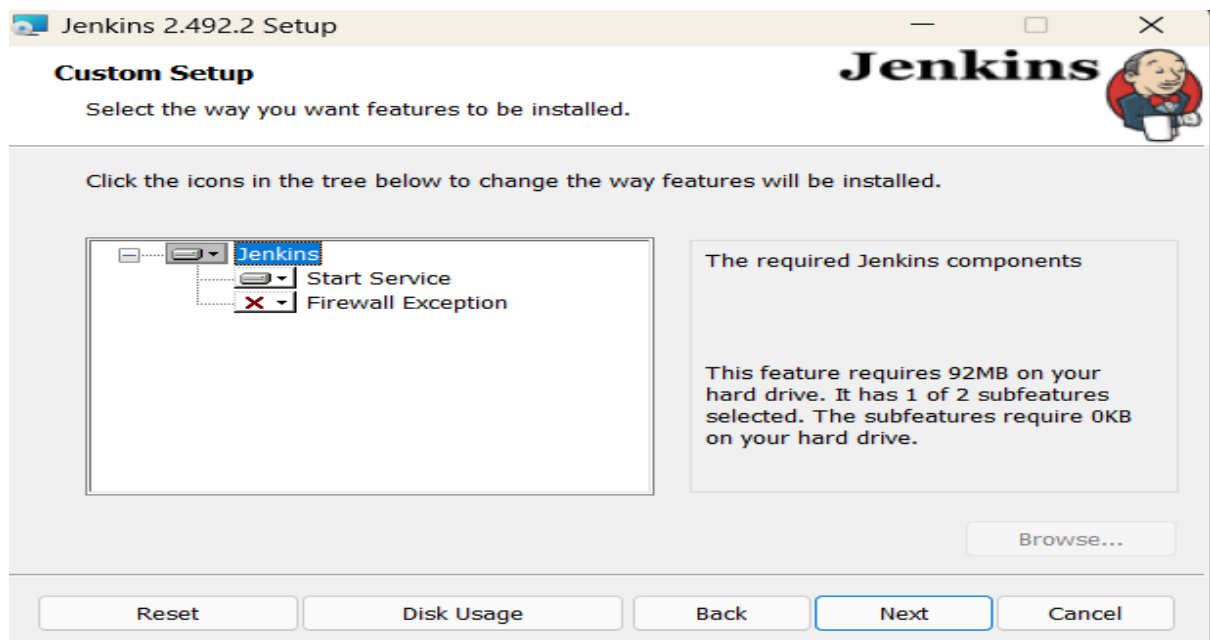


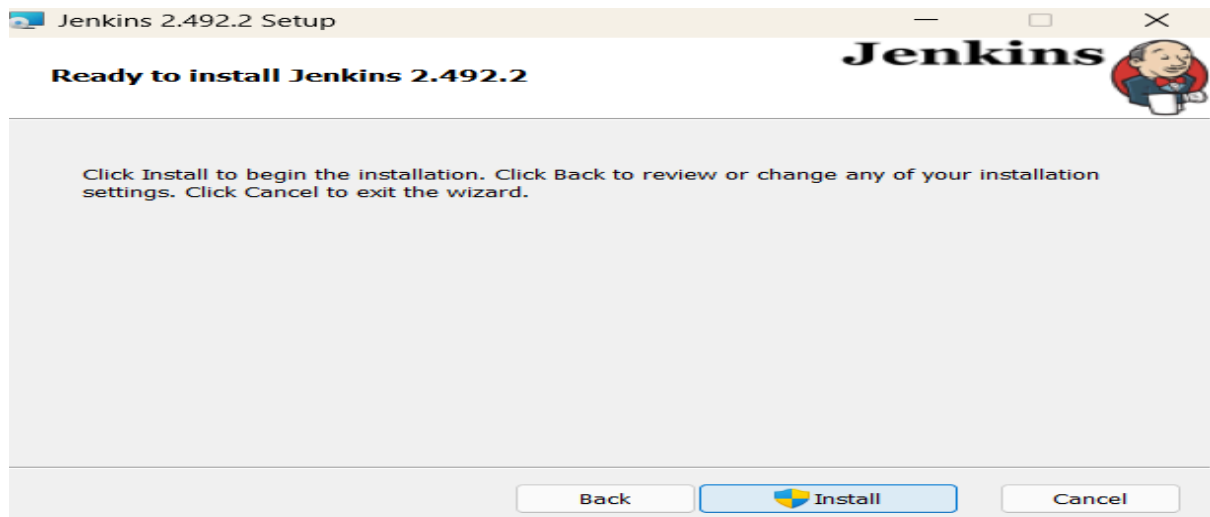


You can see the JDK path here and then click Next



Step 8 :Click Next -----> Click Install -----> Finish

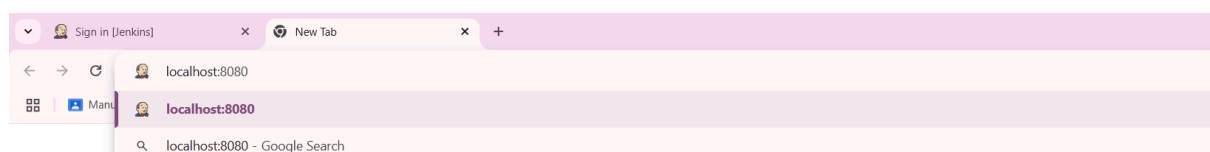




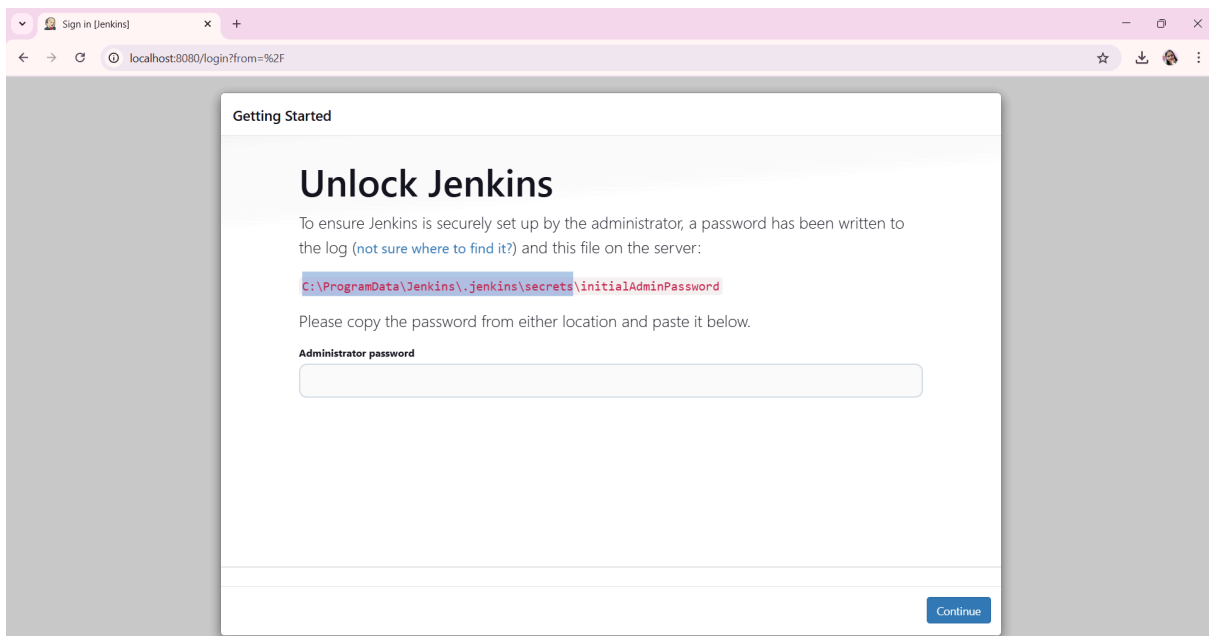
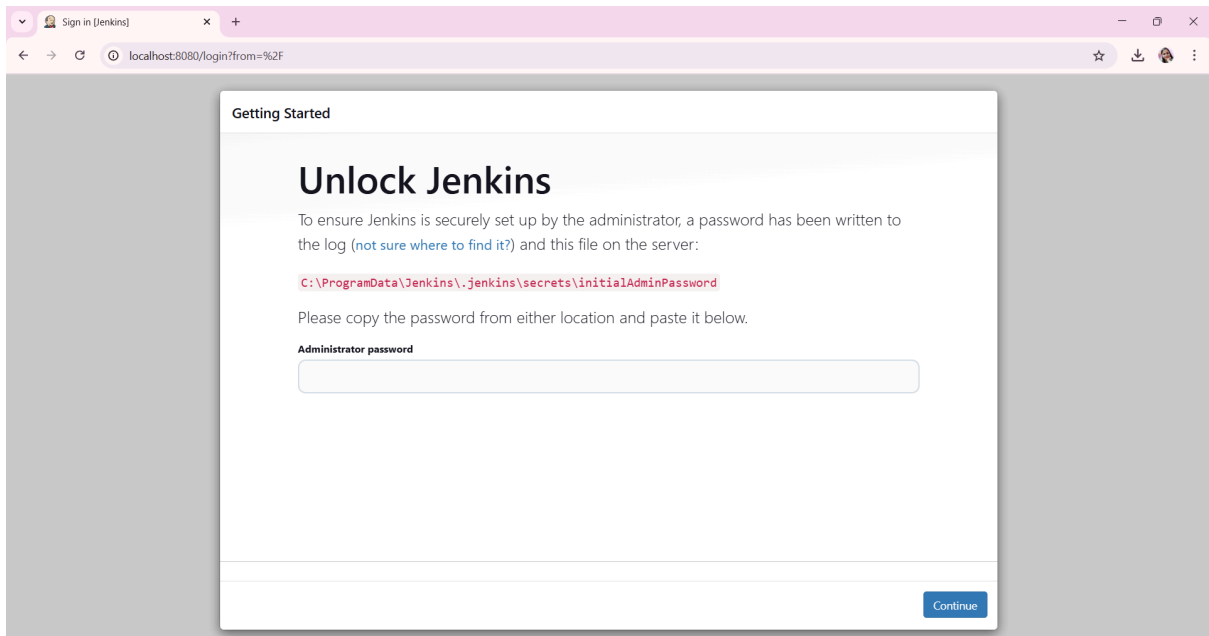
Your have Successfully INSTALLED JENKINS

Step 1: Open a new tab in Chrome -----> in SEARCH BAR type **localhost:8080**
[If you have changed the port number then replace 8080 with you port number]

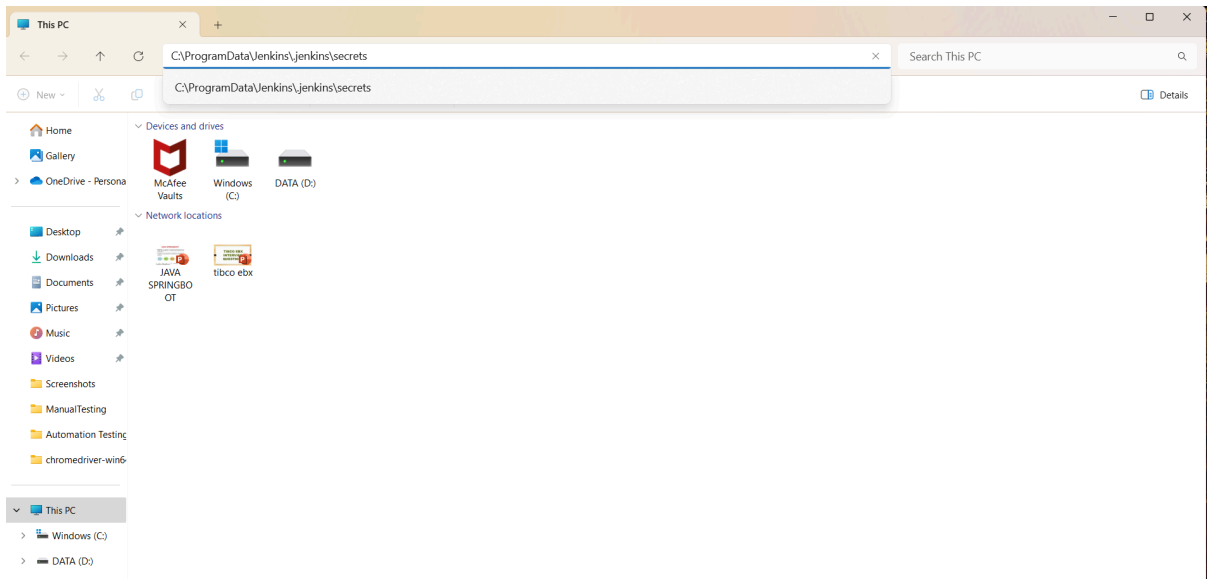
Press ENTER



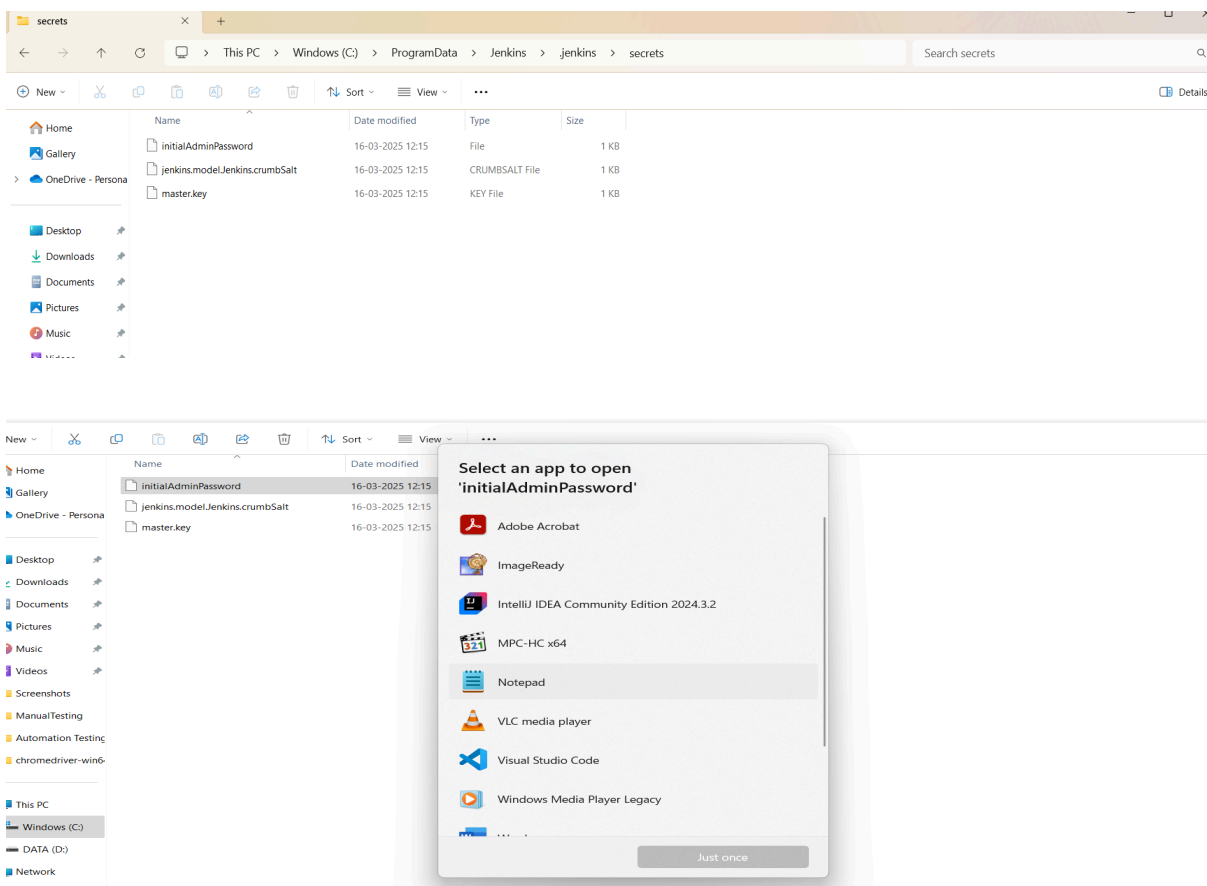
Step 2: Getting Started screen will appear. Here you will see a path, copy that path till SECRETS



Step 3: Go to my computer and paste the path on the top as shown in the image after pasting hit enter



Step 4: In the next screen open initialAdminPassword [Right click and open it in notepad]



Step 5: Copy whatever you got in the notepad and paste it in the Administrative password [In the getting started chrome page] as shown below

And click continue

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

```
C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

.....

Continue

Step 6: In the next page “Customize Jenkins” simply click on install suggested plugins

Getting Started

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

Jenkins 2.492.2

Now it will automatically install required plugins

Step 8: Next Page is the INSTANCE CONFIGURATION

Click on Save and Finish

Getting Started

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.492.2

[Not now](#)

[Save and Finish](#)

Click on Start Using Jenkins

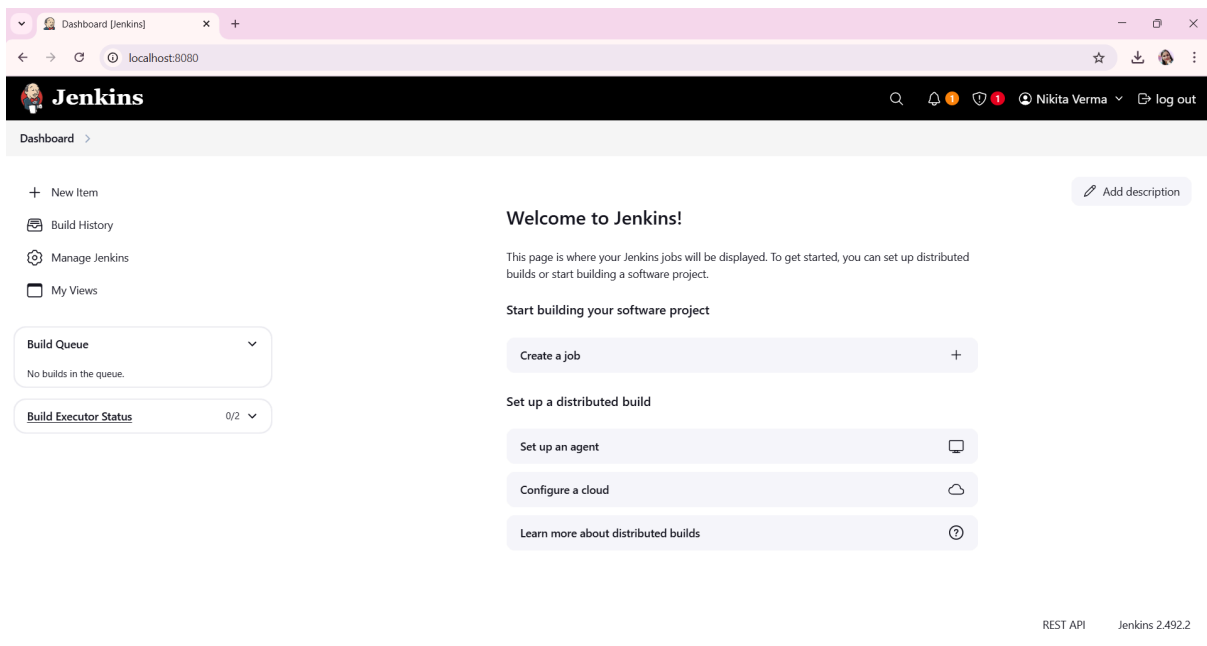
Getting Started

Jenkins is ready!

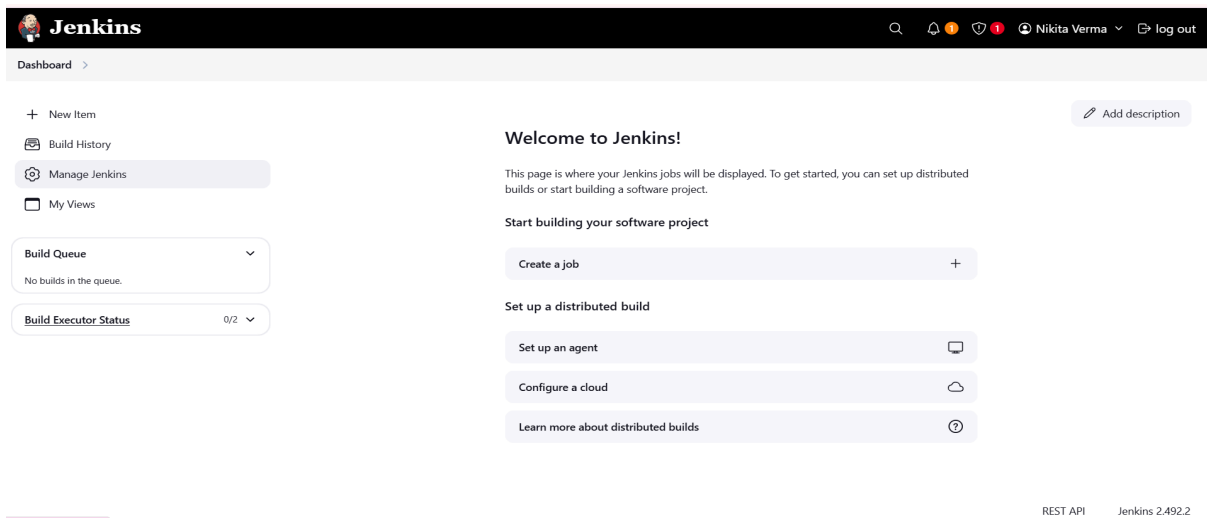
Your Jenkins setup is complete.

[Start using Jenkins](#)

After successfully completing the above steps you will see your JENKINS Dashboard



Step 9: Click On manage jenkins -----> Click On “SYSTEM”

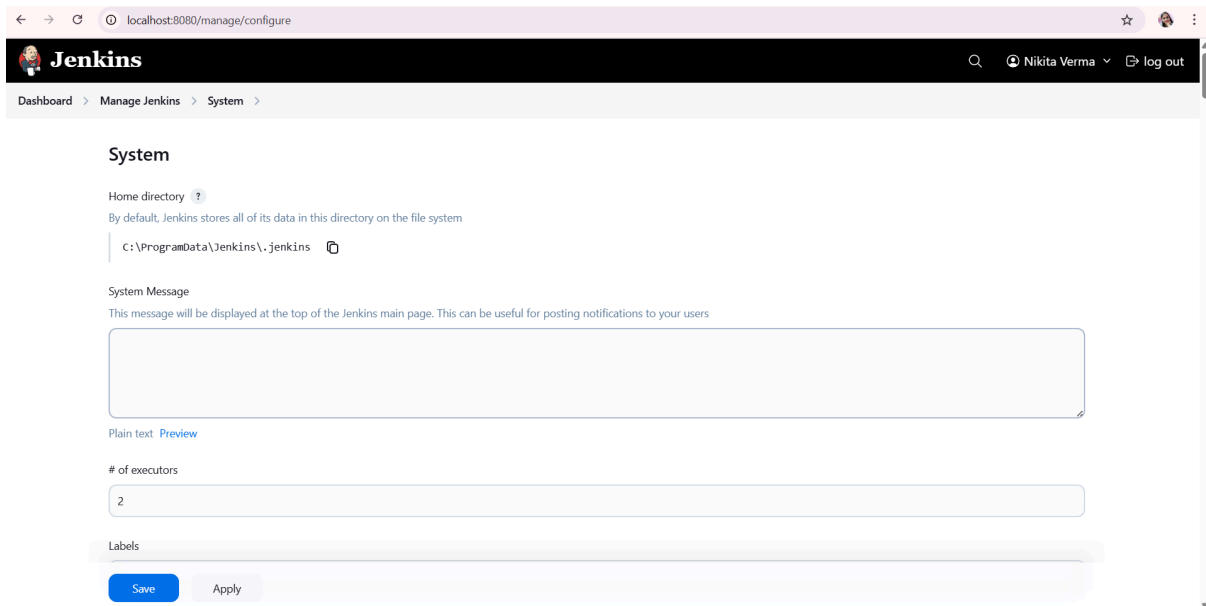


System Configuration

- | | | |
|---|--|---|
|  System
Configure global settings and paths. |  Tools
Configure tools, their locations and automatic installers. |  Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins. |
|  Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on. |  Clouds
Add, remove, and configure cloud instances to provision agents on-demand. |  Appearance
Configure the look and feel of Jenkins |

Security

On the System Page don't change anything, just Click SAVE



System

Home directory ?
By default, Jenkins stores all of its data in this directory on the file system

c:\ProgramData\Jenkins\jenkins

System Message
This message will be displayed at the top of the Jenkins main page. This can be useful for posting notifications to your users

Plain text [Preview](#)

of executors

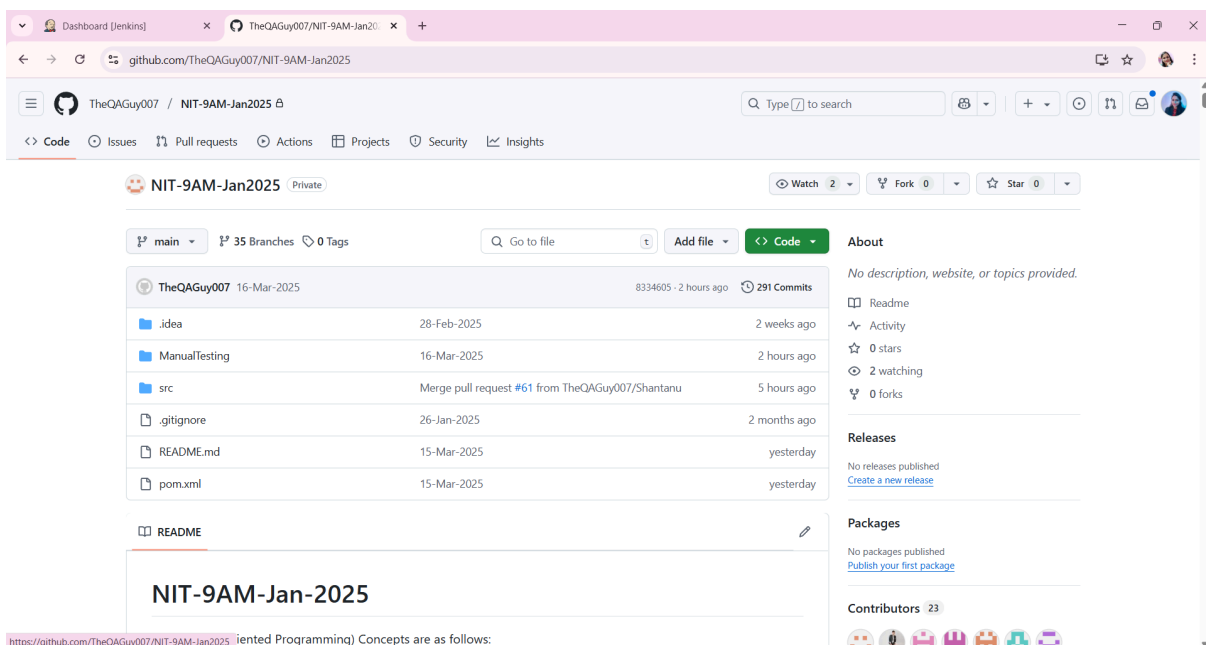
2

Labels

[Save](#) [Apply](#)

HOW TO ADD JDK, GIT AND MAVEN TO JENKINS

Open GITHUB -----> Open NIT Project



Dashboard [Jenkins] x TheQAGuy007/NIT-9AM-Jan2025 x +

github.com/TheQAGuy007/NIT-9AM-Jan2025

TheQAGuy007 / NIT-9AM-Jan2025

Code Issues Pull requests Actions Projects Security Insights

NIT-9AM-Jan2025 Private

Watch 2 Fork 0 Star 0

main 35 Branches 0 Tags

Go to file Add file Code

TheQAGuy007 16-Mar-2025 8334605 · 2 hours ago 291 Commits

idea	28-Feb-2025	2 weeks ago
ManualTesting	16-Mar-2025	2 hours ago
src	Merge pull request #61 from TheQAGuy007/Shantanu	5 hours ago
.gitignore	26-Jan-2025	2 months ago
README.md	15-Mar-2025	yesterday
pom.xml	15-Mar-2025	yesterday

README

NIT-9AM-Jan-2025

About

No description, website, or topics provided.

Readme

Activity

0 stars

2 watching

0 forks

Releases

No releases published

Create a new release

Packages

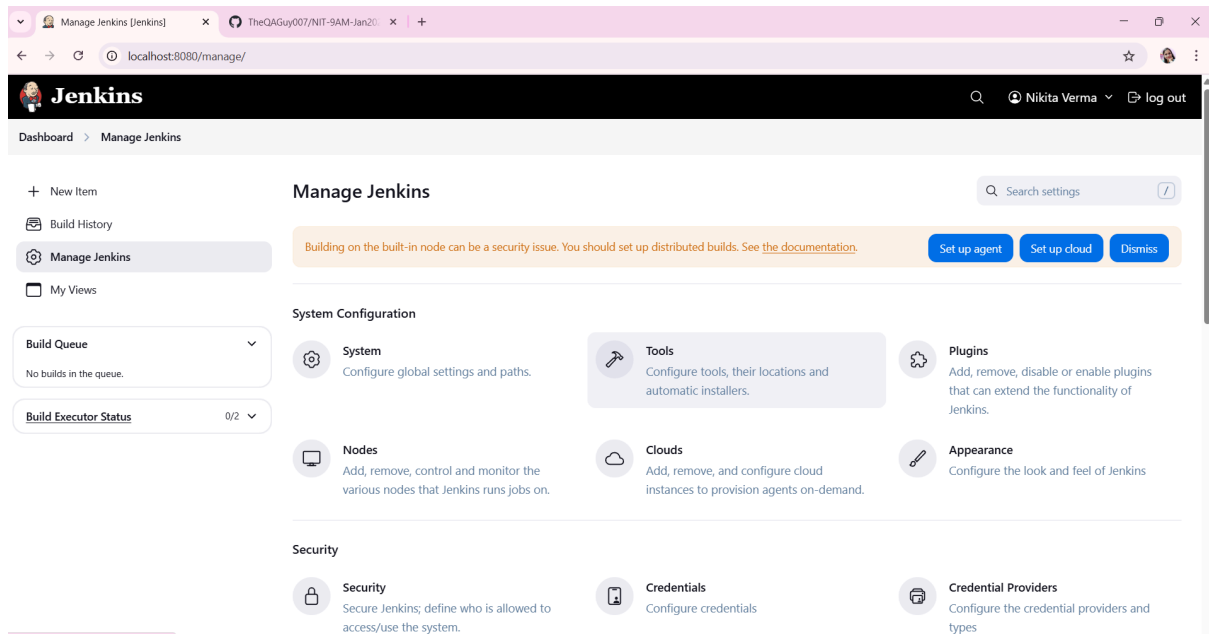
No packages published

Publish your first package

Contributors 23

https://github.com/TheQAGuy007/NIT-9AM-Jan2025 (typed Programming) Concepts are as follows:

Step 1: From JENKINS dashboard Click Manage Jenkins -----> Click TOOLS



Step 2: In tools page Click ADD JDK



In NAME write- JAVA

Dashboard > Manage Jenkins > Tools

JDK installations

[Add JDK](#)

JDK

Name

Required

JAVA_HOME

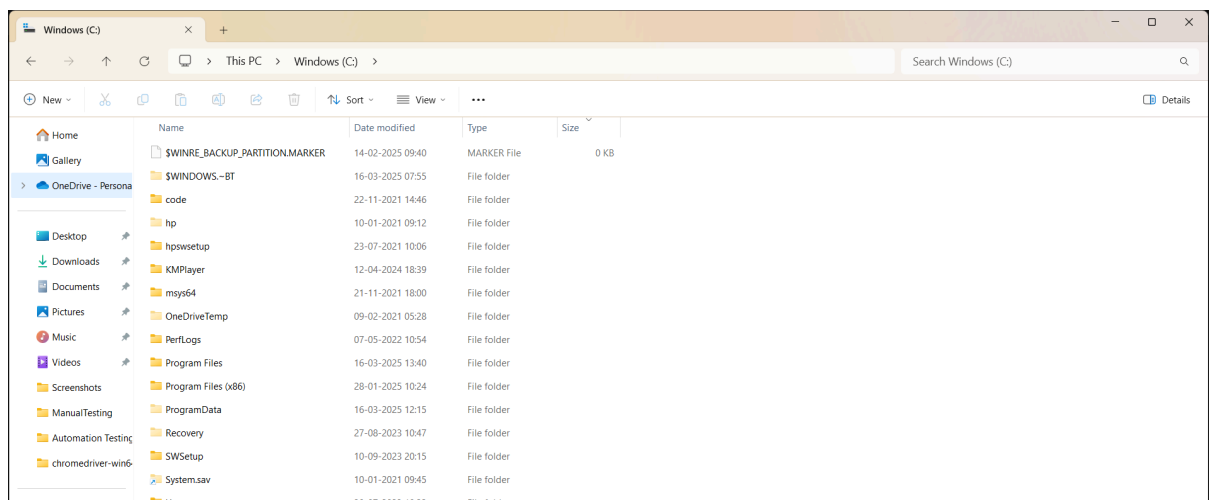
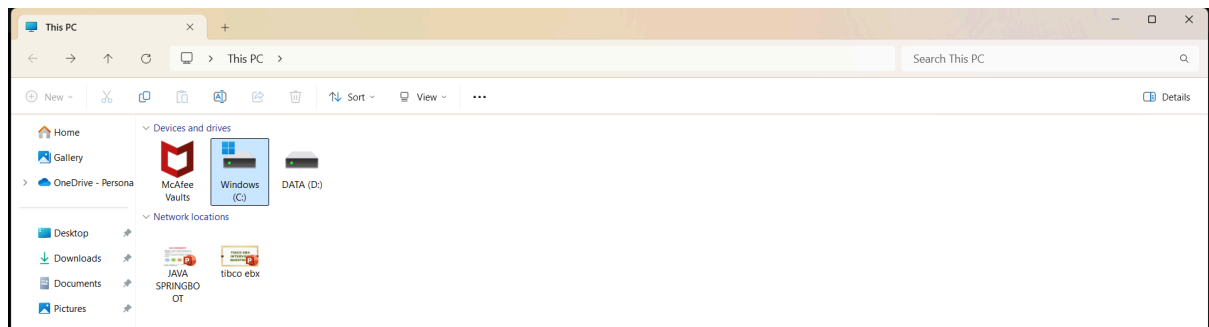
☐ Install automatically ?

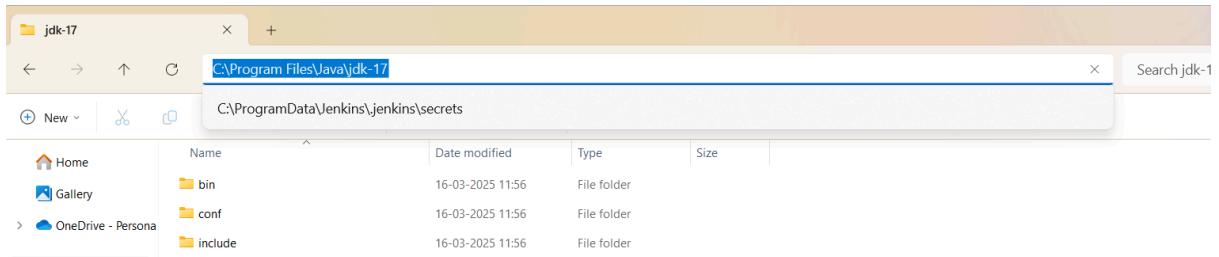
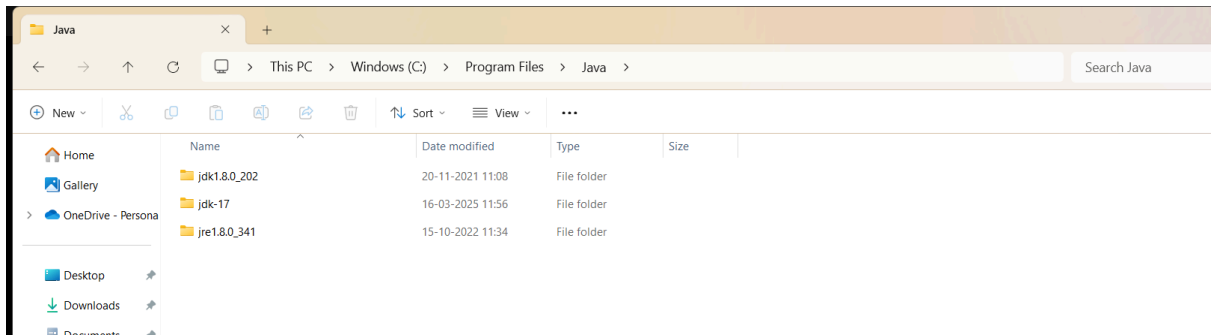
[Add JDK](#)

Git installations

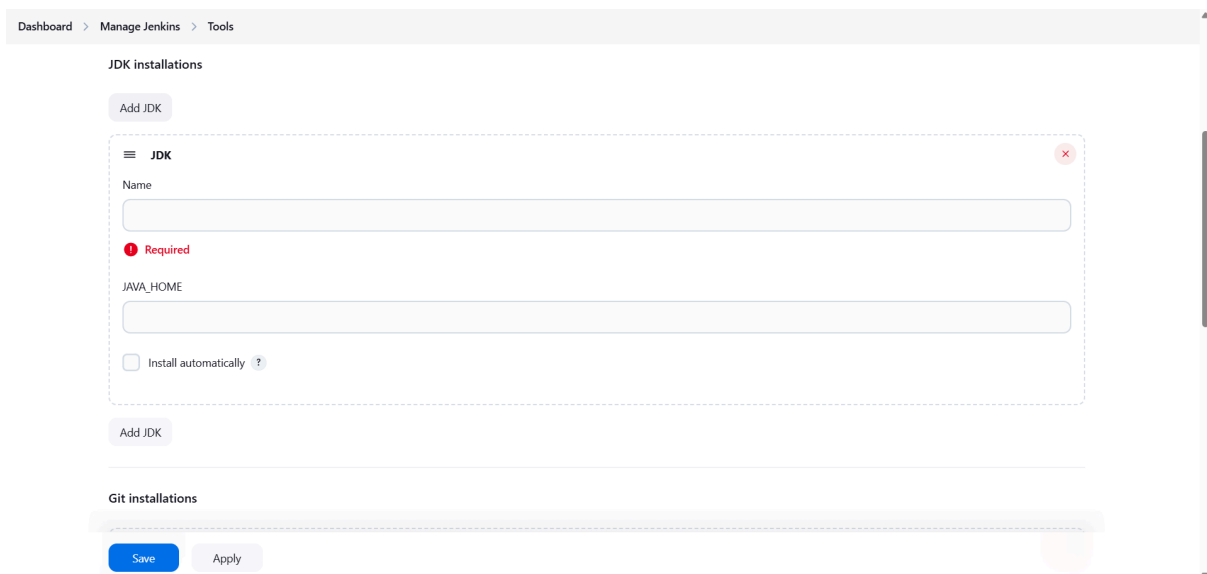
[Save](#) [Apply](#)

Step 3: Go to MY COMPUTER -----> CDRIVE -----> Open Program Files -----> Open JAVA -----> Open JDK folder and then copy the path





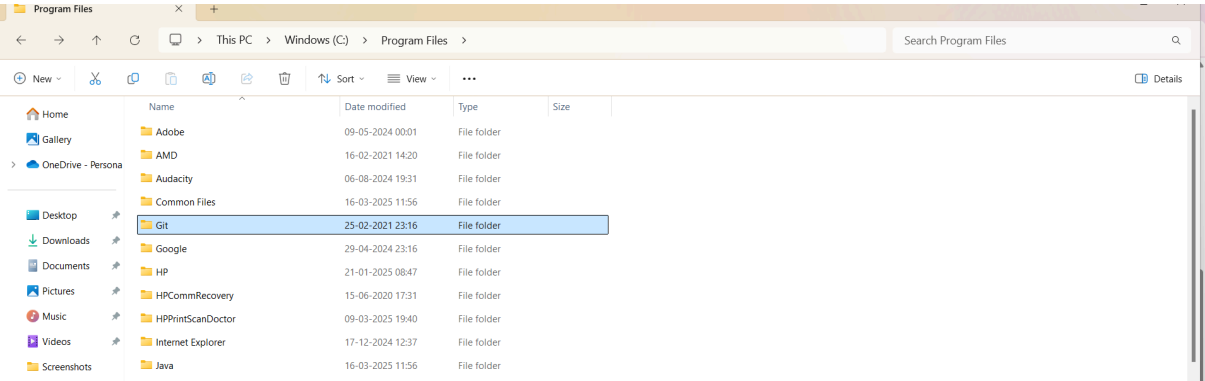
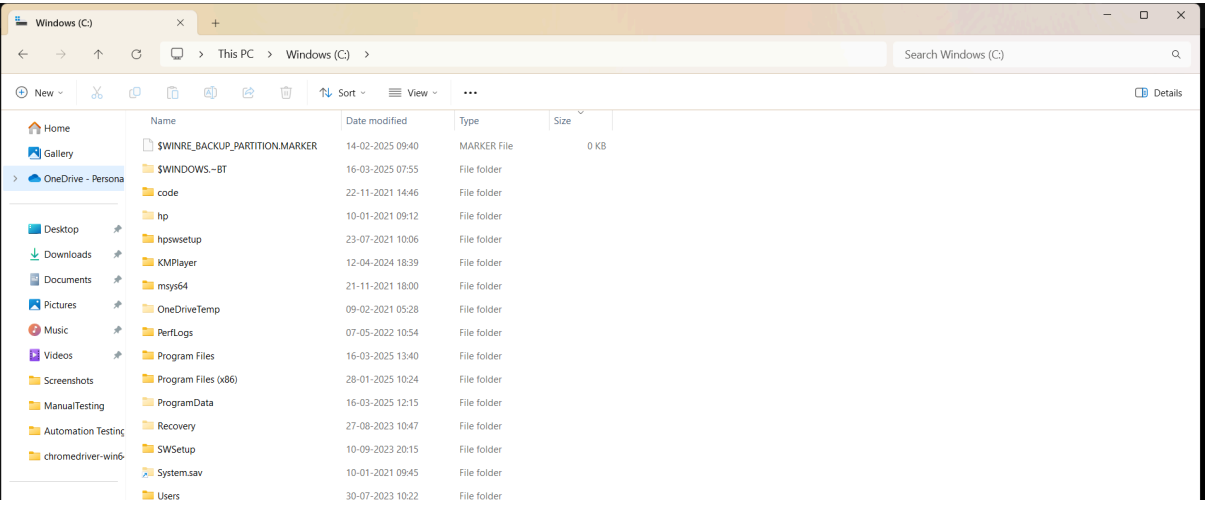
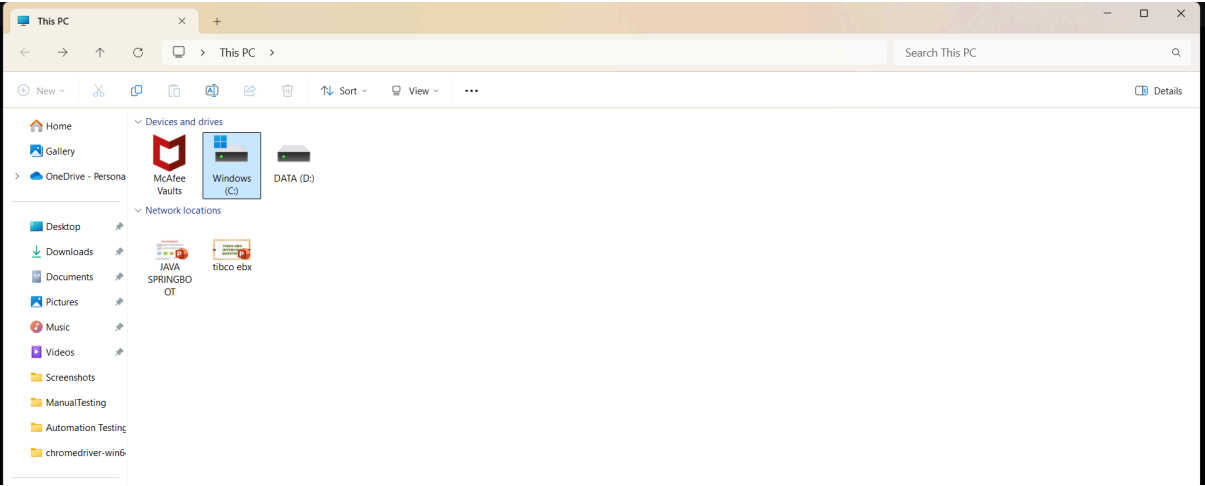
Step 4: Paste this path in JAVA_HOME section in jenkins

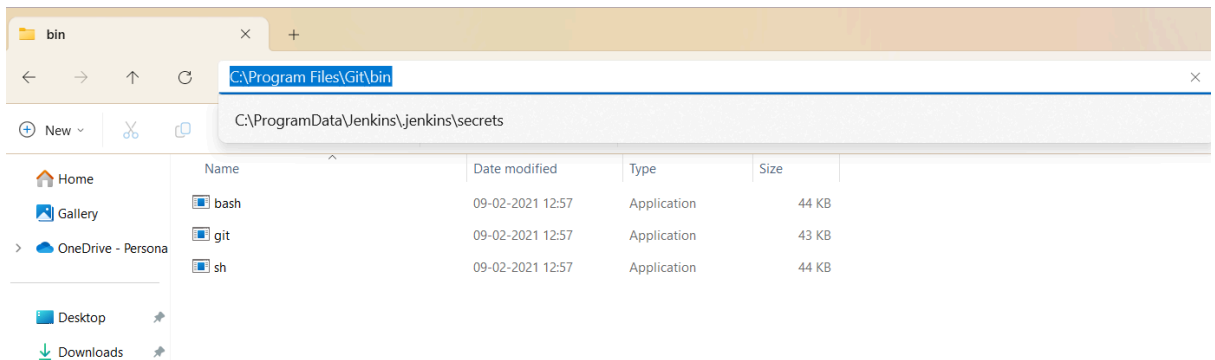
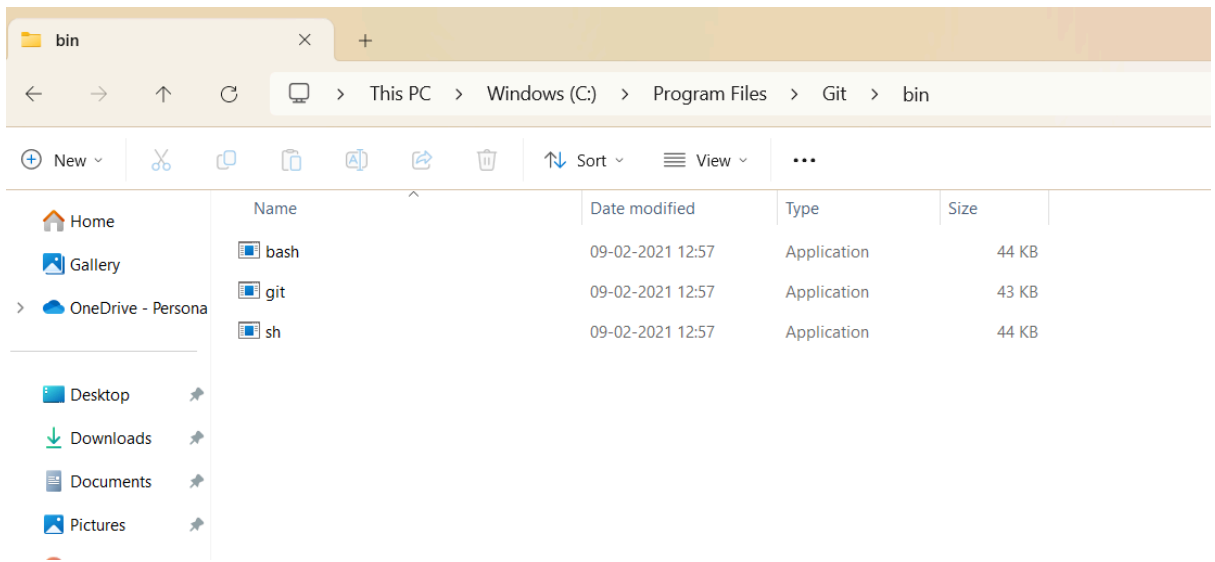
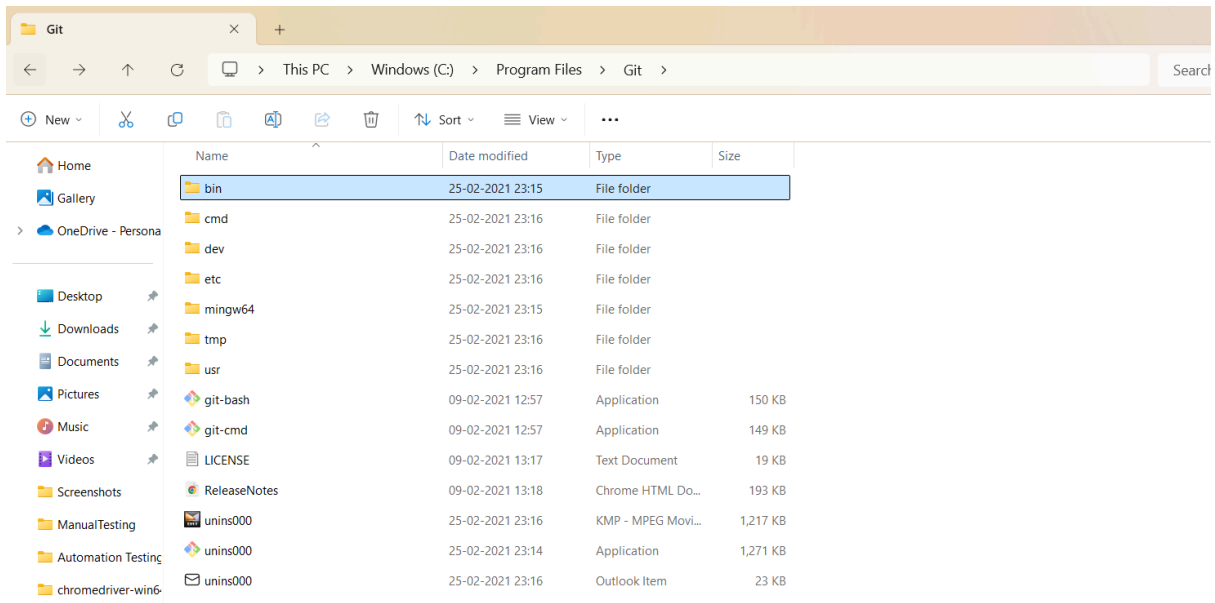


Step 5: Scroll Down and you will see GIT INSTALLATION just below JDK INSTALLATION

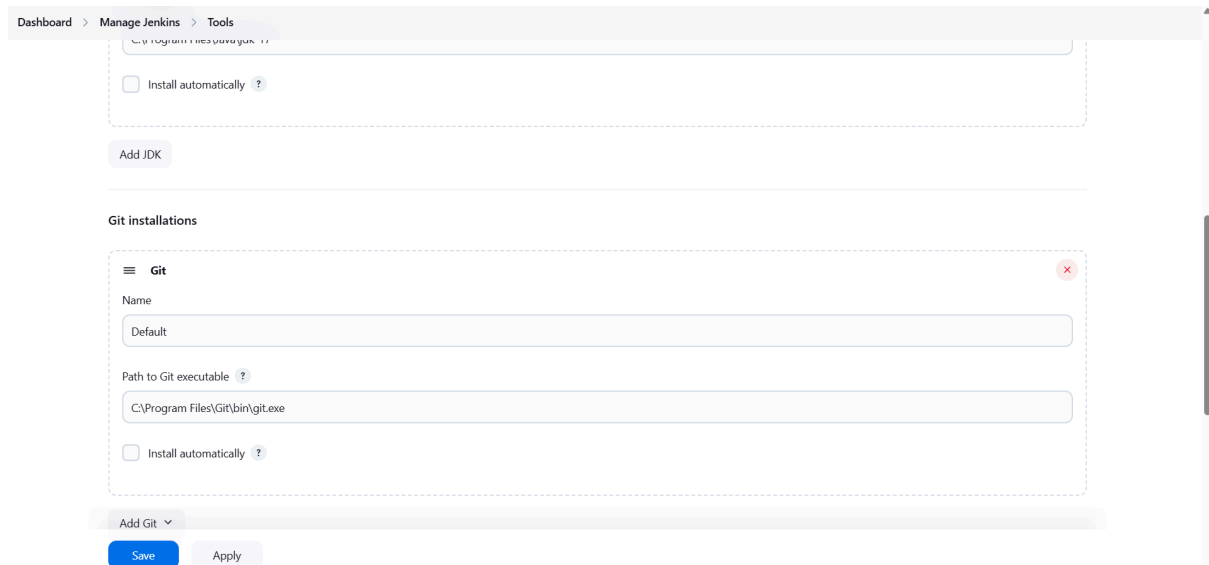
Now to get GIT path :

Go to my computer -----> open Program files -----> Open Git folder -----> Open Bin -----> Copy Path

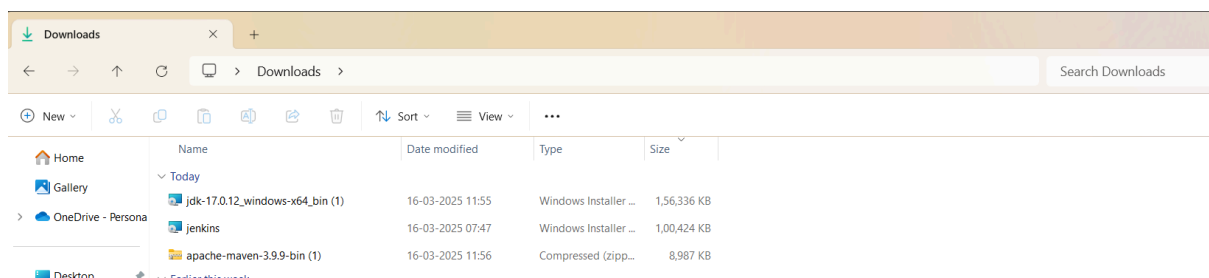




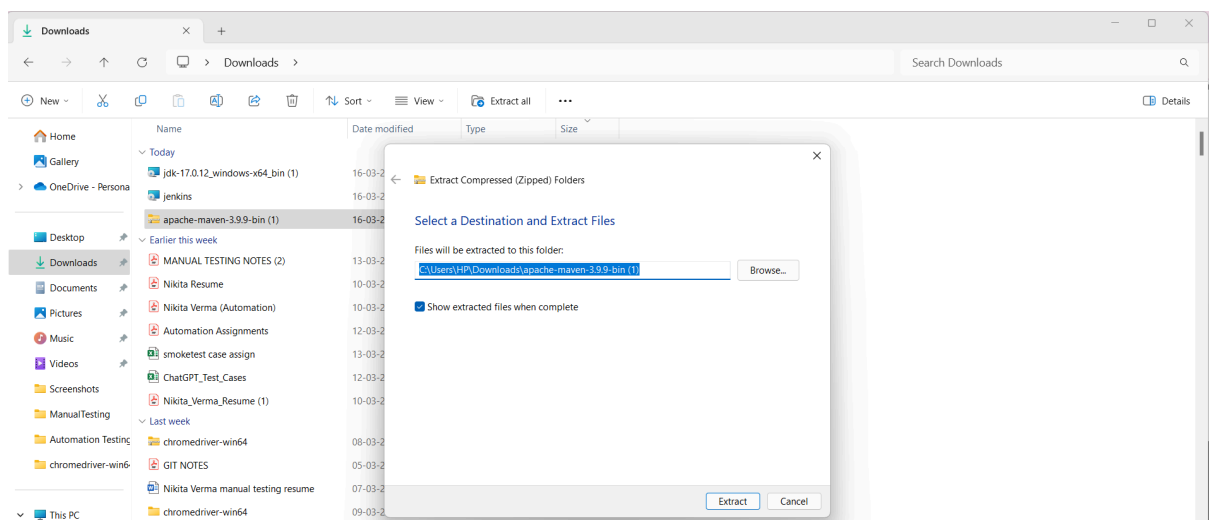
And paste it in the GIT path in jenkins



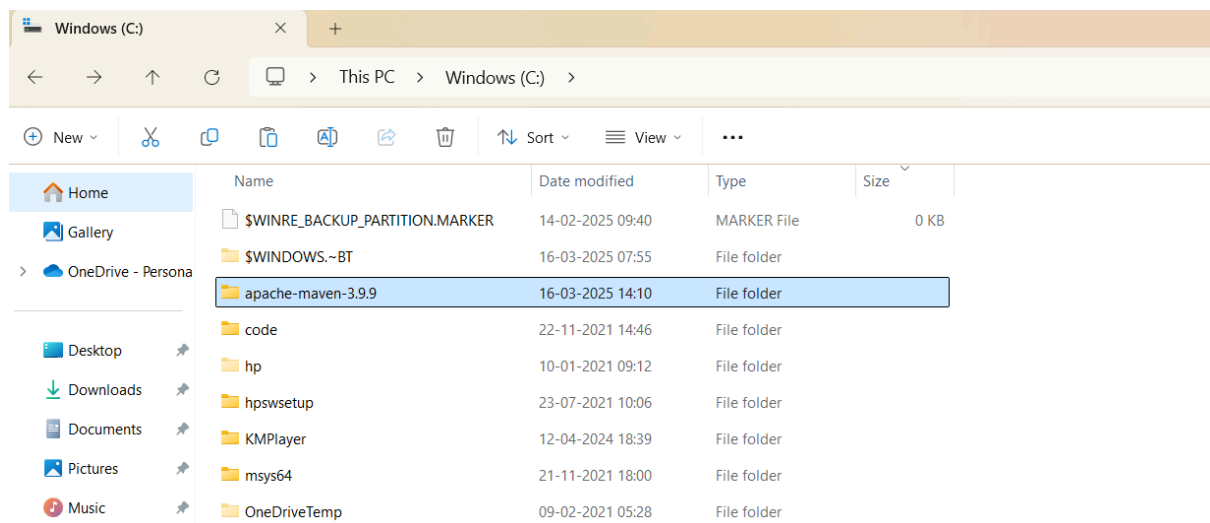
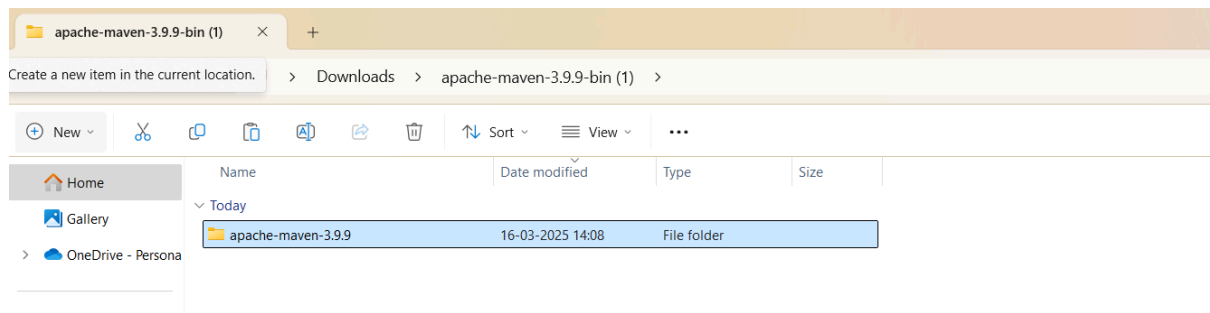
Step 6 : Now Go to MAVEN that you have downloaded
[Maven download link : - <https://maven.apache.org/download.cgi> - Windows Users should download the "apache-maven-3.9.9-bin.zip" file.]



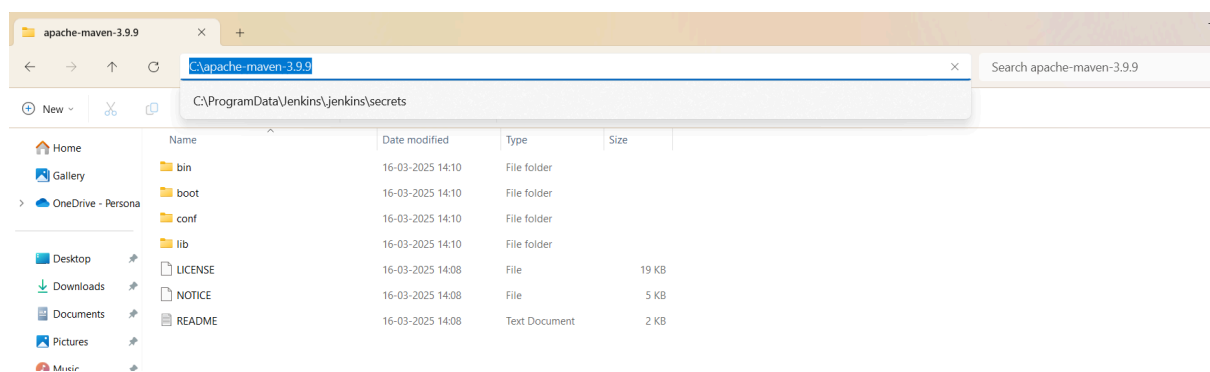
Right Click and extract it



Copy the extracted folder and paste it in the CDRIVE



Step 7: Go inside that folder and copy the path



Step 8: Go back to the same page where you have pasted git nd jdk path, in the same page scroll down you will see MAVEN INSTALLATION, Click ADD MAVEN

Tools [Jenkins] x TheQAGuy007/NIT-9AM-Jan20 x +

localhost:8080/manage/configureTools/

Dashboard > Manage Jenkins > Tools

Add Git

Gradle installations

Add Gradle

Ant installations

Add Ant

Maven installations

Add Maven

Save Apply

In name section write maven
And DESELECT INSTALL AUTOMATICALLY

Dashboard > Manage Jenkins > Tools

Maven

Name

Required

☒ Install automatically ?

Install from Apache

Version

3.9.9

Add Installer

Add Maven

Save Apply

Maven installations

Add Maven

Maven

Name

maven

MAVEN_HOME

☐ Install automatically ?

Add Maven

Save Apply

Then paste the copied path here in MAVEN_HOME

CLICK SAVE AND YOU ARE DONE

Dashboard > Manage Jenkins > Tools

Maven installations

Add Maven

Maven

Name

maven

MAVEN_HOME

C:\apache-maven-3.9.9

☐ Install automatically ?

Add Maven

Save Apply

SECURITY CONFIGURATION

Step 1 : Now Again go back to Dashboard —————> Manage Jenkins —————> And Open Security

localhost:8080/manage/

Dashboard > Manage Jenkins

System Configuration

- Build Queue**
No builds in the queue.
- Build Executor Status** 0/2
- System**
Configure global settings and paths.
- Tools**
Configure tools, their locations and automatic installers.
- Nodes**
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
- Clouds**
Add, remove, and configure cloud instances to provision agents on-demand.
- Plugins**
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
- Appearance**
Configure the look and feel of Jenkins.

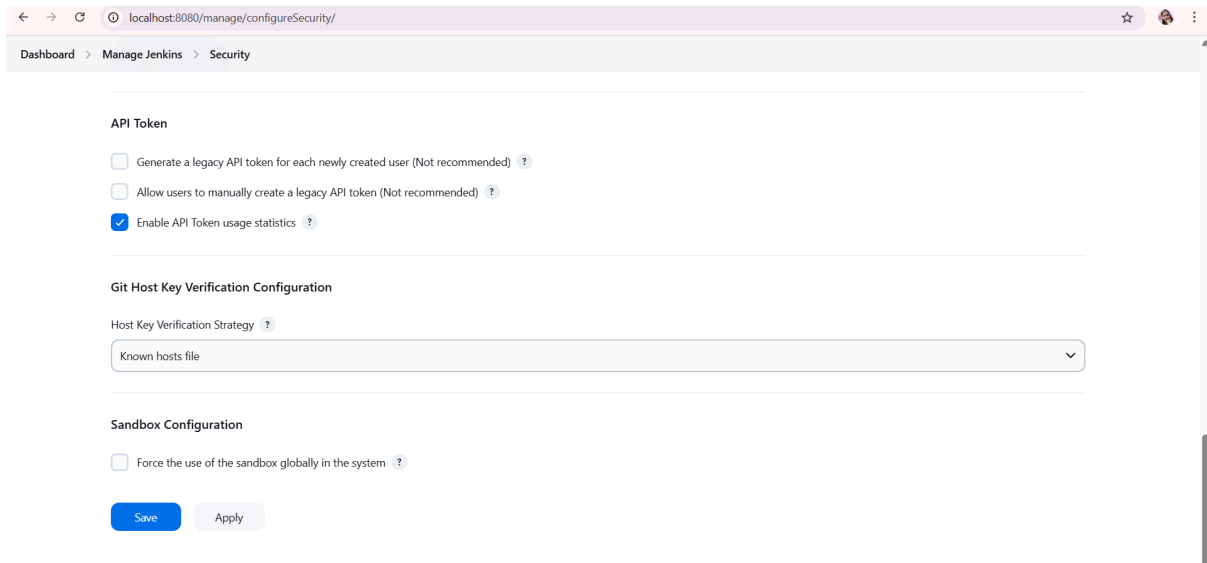
Security

- Security**
Secure Jenkins; define who is allowed to access/use the system.
- Users**
Create/delete/modify users that can log in to this Jenkins.
- Credentials**
Configure credentials.
- Credential Providers**
Configure the credential providers and types.

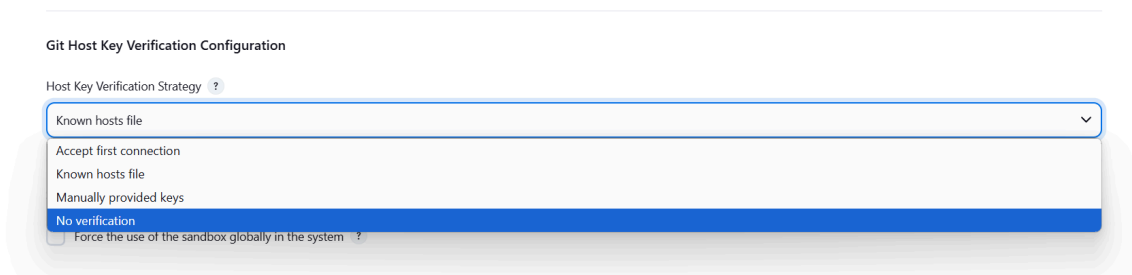
Status Information

- System Information**
Display various environmental
- System Log**
Custom log captures output from
- Load Statistics**
Check your resource utilization and see if

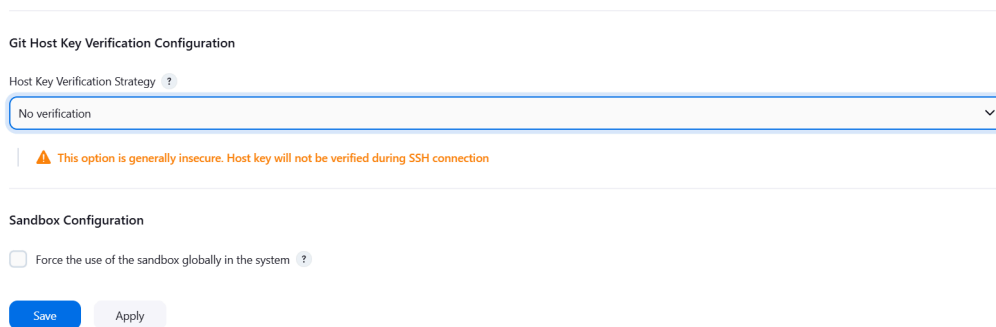
Step 2 : Go to Git host Key Verification Configuration



Here select NO VERIFICATION from drop down



And click SAVE

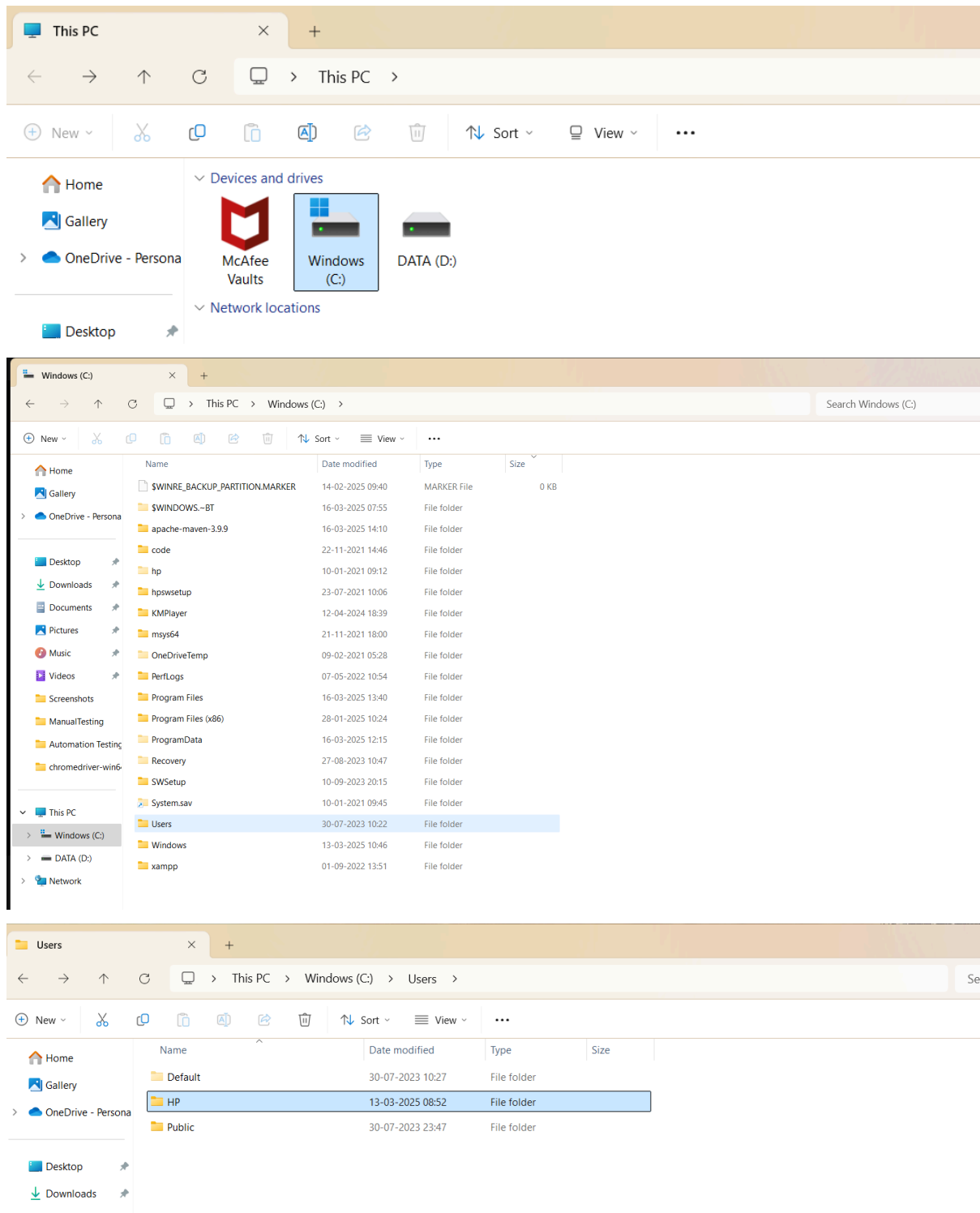


By now all setup are completed and now, we have to CREATE JOBS

INTEGRATING GITHUB WITH JENKINS

SSH KEY GENERATION

Step 1: Go to Cdrive -----> Users ----->And select your Laptop User ----->Open it ----->Go to View-----> Click Show -----> Hidden Items



After unhiding the items, right click Randomly and select OPEN IN TERMINAL
In terminal type **ssh-keygen**
And keep on clicking till you get a triangular pattern

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

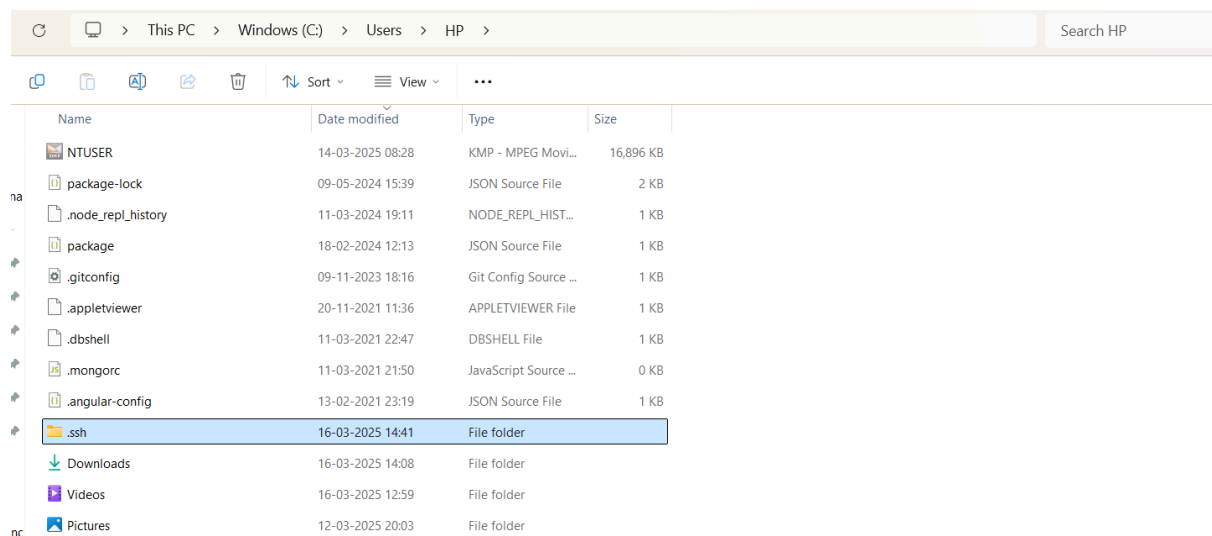
PS C:\Users\HP> ssh-keygen
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

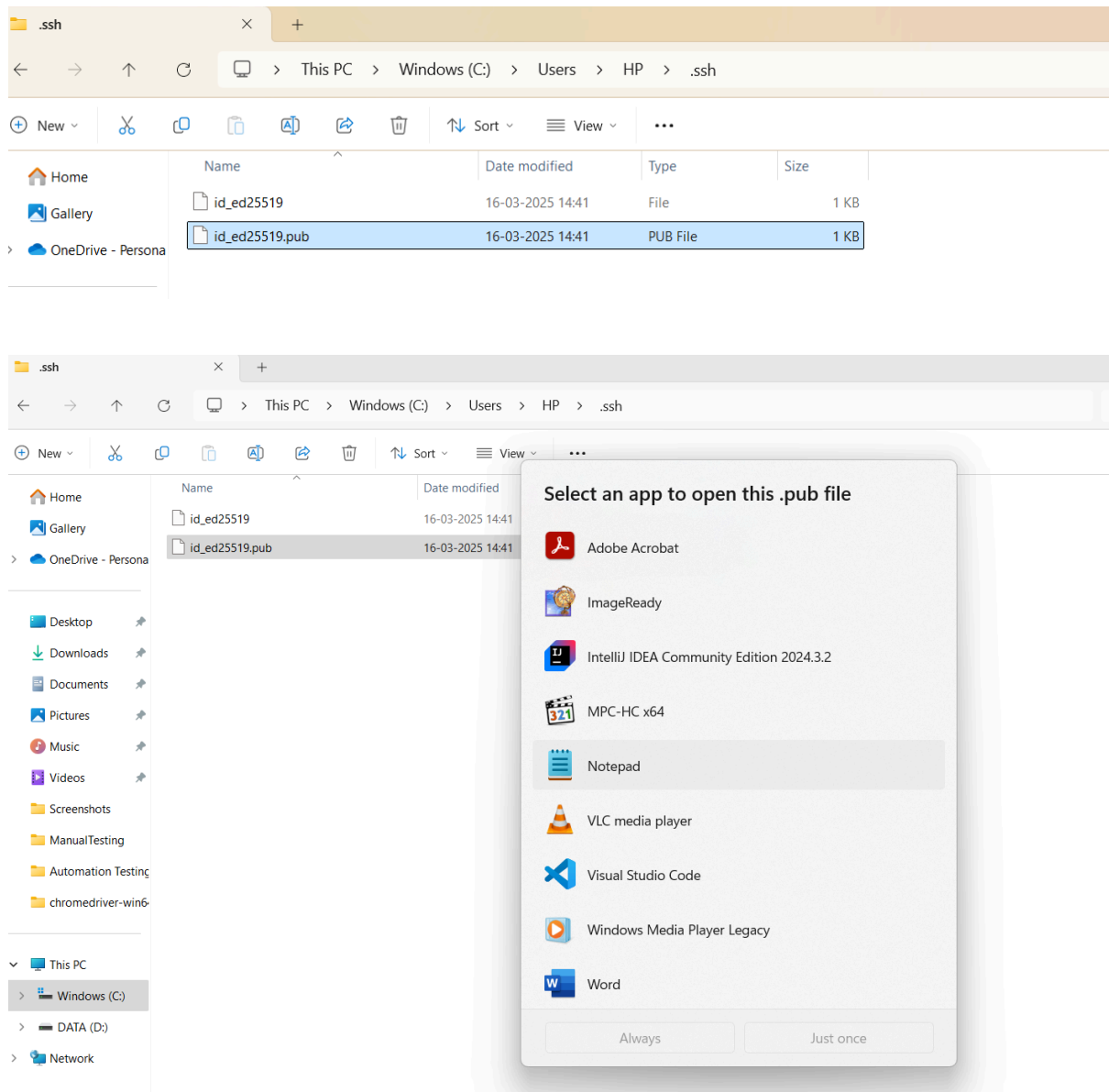
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\HP> ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\HP/.ssh/id_ed25519):
Created directory 'C:\\Users\\HP\\.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\HP/.ssh/id_ed25519
Your public key has been saved in C:\Users\HP/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:TFvvdWgkpR0CxjQ0FiilRk10QsD6EcF6y+NqrcY29UM hp@LAPTOP-QB0532VQ
The key's randomart image is:
+--[ED25519 256]--+
|...  =Bo. . . |
|..o . ..+o . . +|
|o. o o . . . + .|
|o.+ . o o . + . |
|. = o  S . . + . |
|. O. E . . o . |
|.Oo.o . . . |
|.++. o . . |
|=oo . . |
+----[SHA256]-----+
PS C:\Users\HP> |
```

Step 2: Now again go to th User Folder and you will see .ssh folder

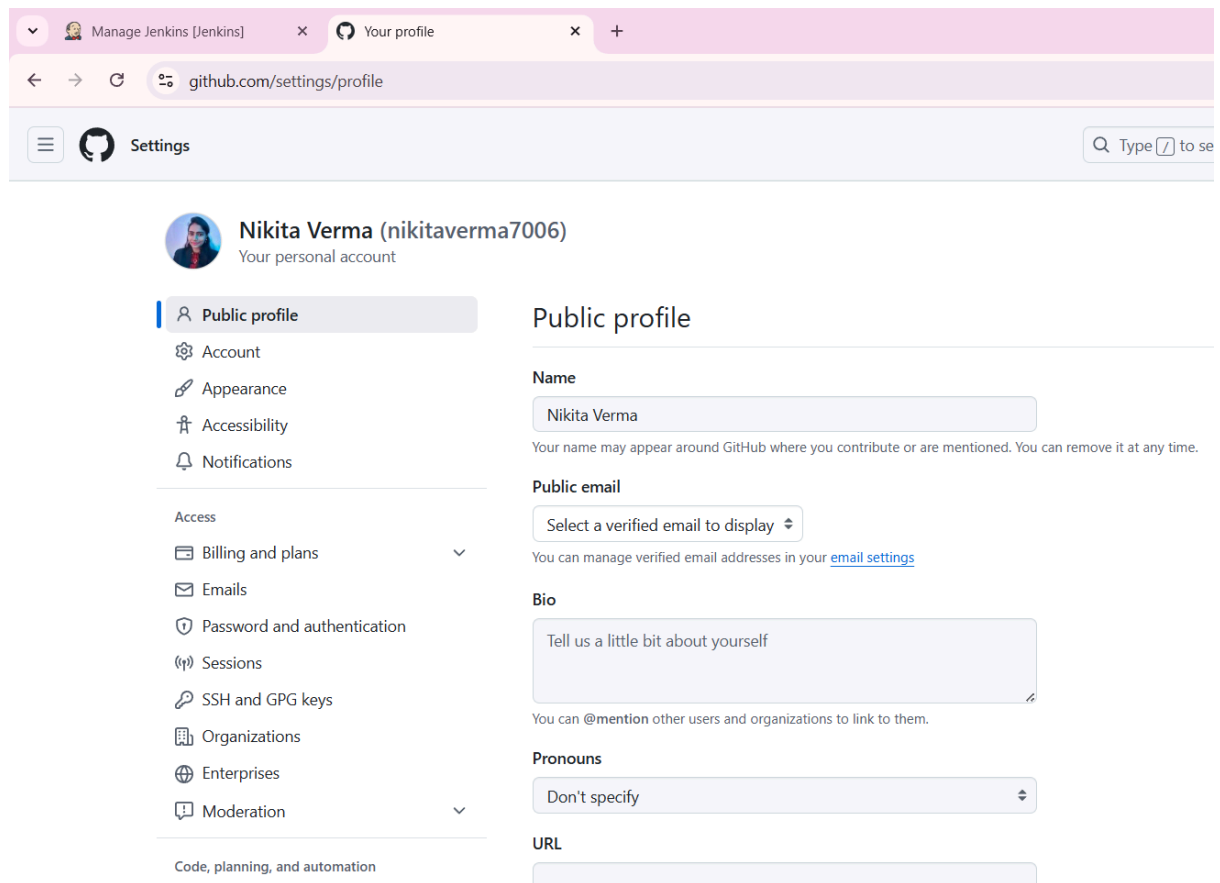


Step 3: Open the .ssh folder-----> Right Click on the **.pub** file and open it in notepad

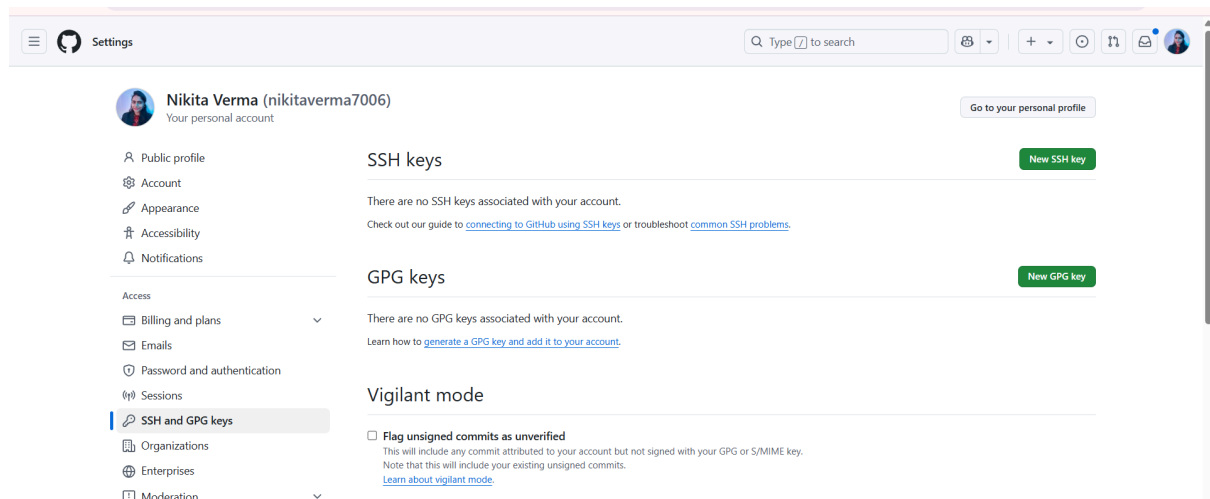


Copy Everything that you got in your notepad

Step 4: Go to GitHub -----> On right side top click on your profile picture icon
-----> Go to settings



Step 5: On left hand side you will see SSH & GPG key , Click it



Step 6: Click On NEW SSH KEY —————> give some title —————> paste that thing that you copied (from the notepad by opening the pub file) in the key section —————> Click ADD SSH KEY
And it's done

Nikita Verma (nikitaverma7006)
Your personal account

Go to your personal profile

Public profile
Account
Appearance
Accessibility
Notifications

Access
Billing and plans
Emails
Password and authentication
Sessions
SSH and GPG keys
Organizations
Enterprises
Moderation

SSH keys

There are no SSH keys associated with your account.
Check out our guide to [connecting to GitHub using SSH keys](#) or troubleshoot [common SSH problems](#).

New SSH key

GPG keys

There are no GPG keys associated with your account.
Learn how to [generate a GPG key and add it to your account](#).

New GPG key

Vigilant mode

☐ **Flag unsigned commits as unverified**
This will include any commit attributed to your account but not signed with your GPG or S/MIME key.
Note that this will include your existing unsigned commits.
[Learn about vigilant mode.](#)

github.com/settings/ssh/new

Nikita Verma (nikitaverma7006)
Your personal account

Go to your personal profile

Public profile
Account
Appearance
Accessibility
Notifications

Access
Billing and plans
Emails
Password and authentication
Sessions
SSH and GPG keys
Organizations
Enterprises
Moderation

Code, planning, and automation
Repositories

Add new SSH Key

Title
keypath

Key type
Authentication Key

Key
ssh-ed25519 AAAAC3NzaC1lZD11NTE5AAAAUQTfK0lB8IBYpZpMWnnQnh1+9/wXSUfh6wPRnns5D hp@LAPTOP-QBO532VQ

Add SSH key

CREDENTIALS Settings

Step1 : GO to the dashboard of JENKINS -----> Manage Jenkins -----> Go to Credentials

localhost:8080/manage/

Dashboard > Manage Jenkins

Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

Clouds
Add, remove, and configure cloud instances to provision agents on-demand.

Appearance
Configure the look and feel of Jenkins

Security

Security
Secure Jenkins; define who is allowed to access/use the system.

Users
Create/delete/modify users that can log in to this Jenkins.

Credentials
Configure credentials

Credential Providers
Configure the credential providers and types

Status Information

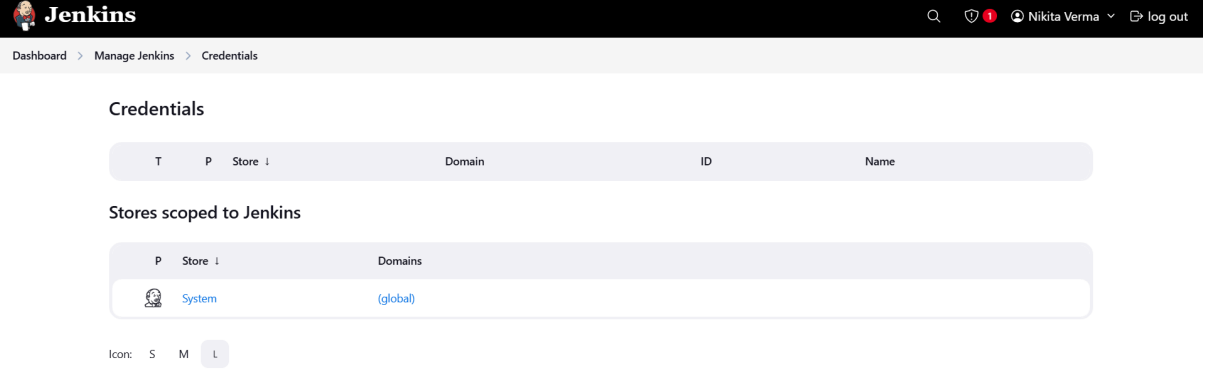
System Information
Displays various environmental information to assist trouble-shooting.

System Log
System log captures output from java.util.logging output related to Jenkins.

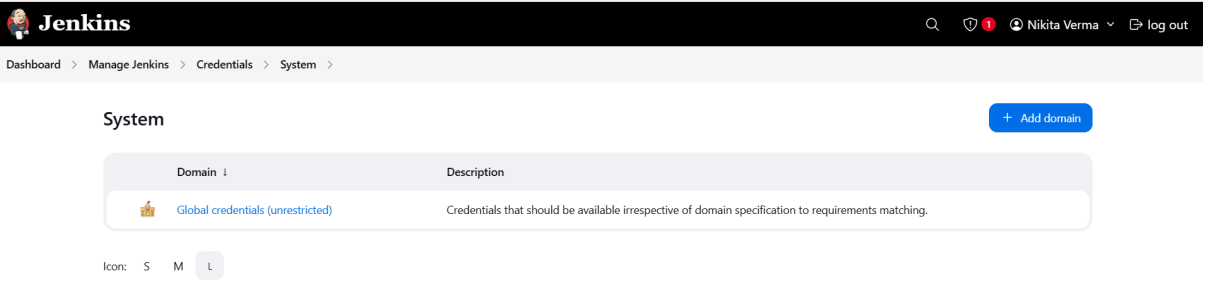
Load Statistics
Check your resource utilization and see if you need more computers for your builds.

About Jenkins
See the version and license information.

Step 2 : Go to System -----> Global Credentials

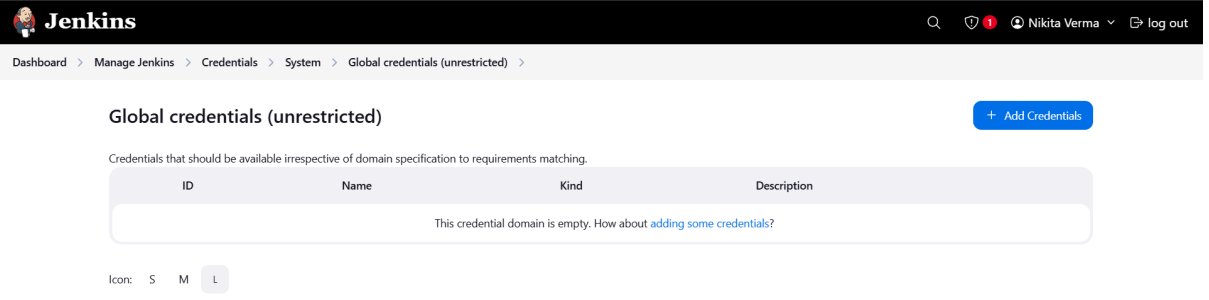


The screenshot shows the Jenkins 'Credentials' page. The breadcrumb trail is 'Dashboard > Manage Jenkins > Credentials'. The page title is 'Credentials'. Below the title is a table with columns: 'T', 'P', 'Store', 'Domain', 'ID', and 'Name'. Under the heading 'Stores scoped to Jenkins', there is a table with columns 'P', 'Store', and 'Domains'. It lists one entry: 'System' with a '(global)' domain. At the bottom, there are icons for 'S', 'M', and 'L', with 'L' being selected.



The screenshot shows the Jenkins 'System' page. The breadcrumb trail is 'Dashboard > Manage Jenkins > Credentials > System'. The page title is 'System'. There is a '+ Add domain' button. Below is a table with columns 'Domain' and 'Description'. It lists one entry: 'Global credentials (unrestricted)' with a description: 'Credentials that should be available irrespective of domain specification to requirements matching.' At the bottom, there are icons for 'S', 'M', and 'L', with 'L' being selected.

Step 3: Click On ADD CREDENTIALS



The screenshot shows the Jenkins 'Global credentials (unrestricted)' page. The breadcrumb trail is 'Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted)'. The page title is 'Global credentials (unrestricted)'. There is a '+ Add Credentials' button. Below the title is a message: 'Credentials that should be available irrespective of domain specification to requirements matching.' Below this is a table with columns: 'ID', 'Name', 'Kind', and 'Description'. The table is empty, and a message at the bottom says: 'This credential domain is empty. How about [adding some credentials?](#)' At the bottom, there are icons for 'S', 'M', and 'L', with 'L' being selected.

Step 4 : Now in credentials Page

In KIND choose "SSH USER NAME WITH PRIVATE KEY"

In ID - give your Github user name

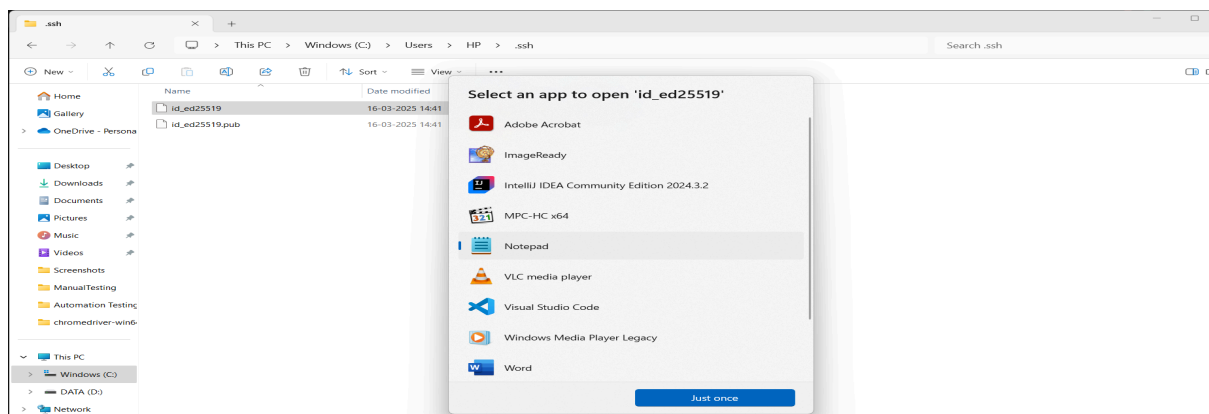
In Username - write the same githubhub user name

In private key - Select Enter Directly

The screenshot shows the Jenkins 'New credentials' form. The 'Kind' is set to 'Username with password'. The 'Scope' is 'Global (Jenkins, nodes, items, all child items, etc)'. The 'Username' and 'Password' fields are empty. The 'ID' field is 'nikitaverma7006'. The 'Description' field is empty. The 'Username' field is 'nikitaverma7006'. The 'Private Key' section has 'Enter directly' selected. The 'Key' field is empty. The 'Passphrase' field is empty. The 'Create' button is visible at the bottom.

Step 5 : Now go back to the SSH Folder in Cdriver and copy the other file data and paste it in KEY SECTION

And Click CREATE



Username

nikitaverma7006

☐ Treat username as secret ?

Private Key

☒ Enter directly

Key

Enter New Secret Below

```
-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnIzZaC1rZXktZjEAAAAAG5vbmUAAAAAEbm9uZQAAAAAAAAABAAAAMwAAATzc2gtZW
OyNTUxO0AAACCUEXSTCGxYPjHd8S6TFp50J4Zfvf8F01H4esD6Z5700wAAA3hu2uP9BtrT
```

Passphrase

Create

REST API Jenkins 2.492.2

And your GLOBAL CREDENTIALS is CREATED

localhost:8080/manage/credentials/store/system/domain/_/

Jenkins

Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted) >

Global credentials (unrestricted) [+ Add Credentials](#)

Credentials that should be available irrespective of domain specification to requirements matching.

ID	Name	Kind	Description
nikitaverma7006	nikitaverma7006	SSH Username with private key	

Icon: S M L

CREATING JOBS IN JENKINS

Step 1 : GO to Jenkins DASHBOARD -----> Click New Items -----> Enter an item name -----> Click on FreeStyle Project -----> Click Ok

Jenkins

Dashboard >

+ New Item

Build History

Manage Jenkins

My Views

Build Queue

No builds in the queue.

Build Executor Status

0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

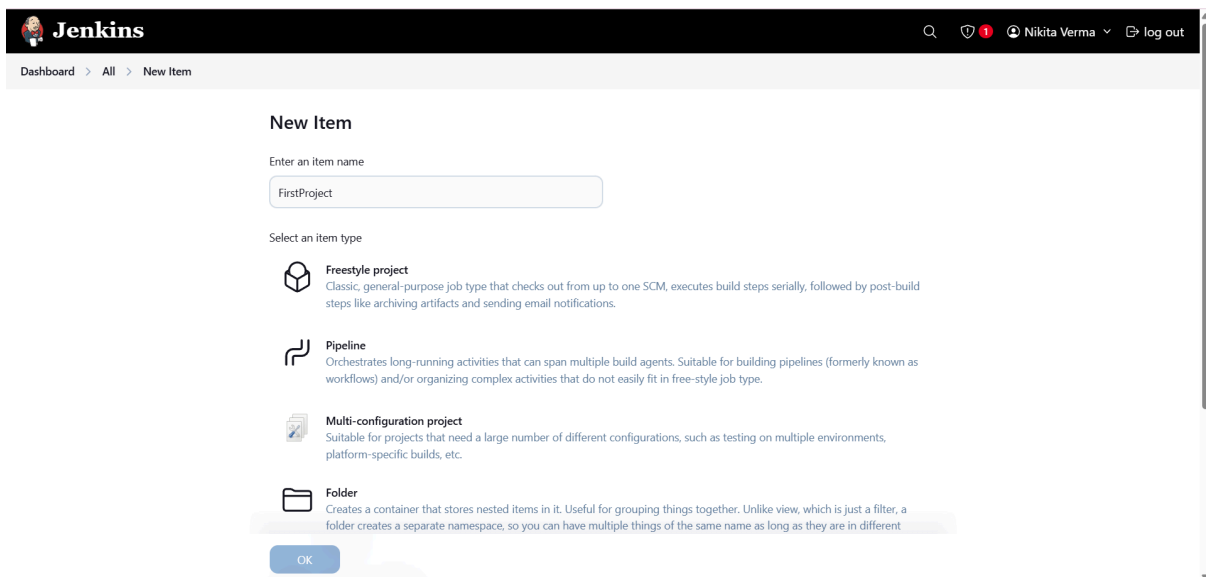
Create a job

Set up a distributed build

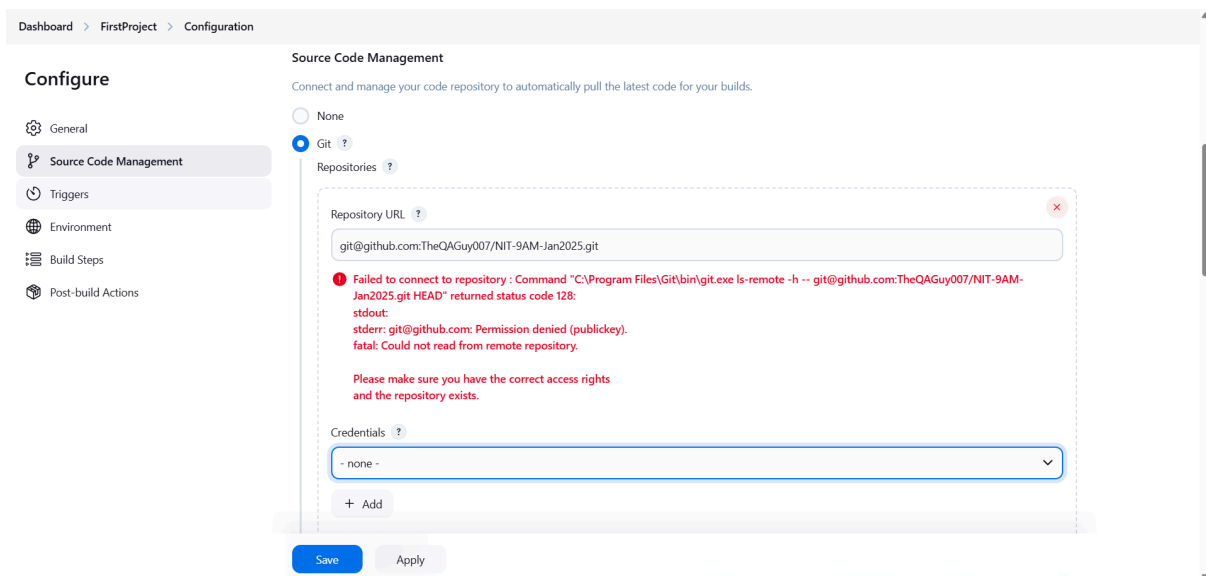
Set up an agent

Configure a cloud

Learn more about distributed builds



Step 2: Go to your Job that you have created now , In the “Source Code Management” —————. Select Git



Step 3 : Go to Your Project in GitHub —————> Click on the Green colour Code Button present under NIT-9AM-JAN2025

There Select “SSH” and then copy it and Paste it in The REPOSITORY URL of Jenkins

TheQAGuy007 / NIT-9AM-Jan2025

Type to search

<> Code

Issues

Pull requests

Actions

Projects

Security

Insights

NIT-9AM-Jan2025

Watch2Fork0Star0

main35 Branches0 Tags

Go to file

Add file

Code

TheQAGuy00716-Mar-20258334605 · 4 hours ago291 Commits

idea	28-Feb-2025	2 weeks ago
ManualTesting	16-Mar-2025	4 hours ago
src	Merge pull request #61 from TheQAGuy007/Shantanu	7 hours ago
.gitignore	26-Jan-2025	2 months ago
README.md	15-Mar-2025	yesterday
pom.xml	15-Mar-2025	yesterday

README

About

No description, website, or topics provided.

Readme

Activity

0 stars

2 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors23

TheQAGuy007 / NIT-9AM-Jan2025

Type to search

<> Code

Issues

Pull requests

Actions

Projects

Security

Insights

NIT-9AM-Jan2025

Watch2Fork0Star0

main35 Branches0 Tags

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Add file

Code

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.gitignore	26-Jan-2025	2 months ago
README.md	15-Mar-2025	yesterday
pom.xml	15-Mar-2025	yesterday

README

About

No description, website, or topics provided.

Readme

Activity

0 stars

2 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Local

Codespaces

Clone

HTTPSSSHGitHub CLI

git@github.com:TheQAGuy007/NIT-9AM-Jan2025.git

Use a password-protected SSH key.

Open with GitHub Desktop

Download ZIP

Dashboard > FirstProject > Configuration

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

None

Git

Repositories

Repository URL

git@github.com:TheQAGuy007/NIT-9AM-Jan2025.git

Failed to connect to repository : Command "C:\Program Files\Git\bin\git.exe ls-remote -h -- git@github.com:TheQAGuy007/NIT-9AM-Jan2025.git HEAD" returned status code 128:

stdout:

stderr: git@github.com: Permission denied (publickey).

fatal: Could not read from remote repository.

Please make sure you have the correct access rights and the repository exists.

Credentials

- none -

+ Add

Save

Apply

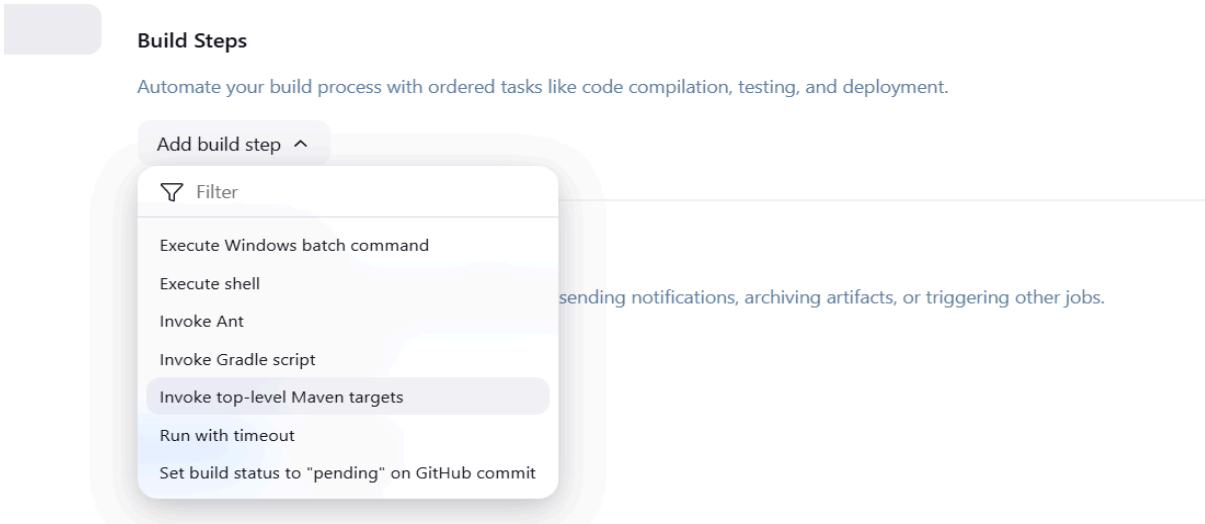
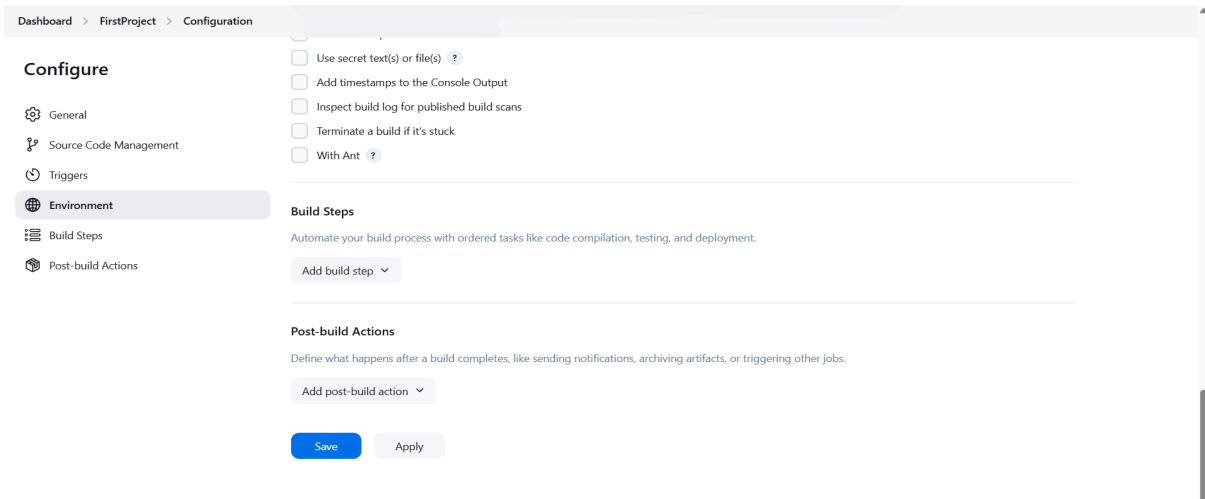
Step 4 : Select Proper Credentials and the error will be gone (will take few seconds)

The screenshot shows the 'Source Code Management' configuration page. On the left, a sidebar lists configuration options: General, Source Code Management (selected), Triggers, Environment, Build Steps, and Post-build Actions. The main area is titled 'Source Code Management' and includes a sub-header 'Connect and manage your code repository to automatically pull the latest code for your builds.' Below this, there are radio buttons for 'None' and 'Git' (selected). A 'Repositories' section contains a 'Repository URL' field with the value 'git@github.com:TheQAGuy007/NIT-9AM-Jan2025.git', a 'Credentials' dropdown menu showing 'nikitavarma7006', and an 'Add' button. There is also an 'Advanced' dropdown and an 'Add Repository' button. At the bottom, there is a 'Branches to build' section and 'Save' and 'Apply' buttons.

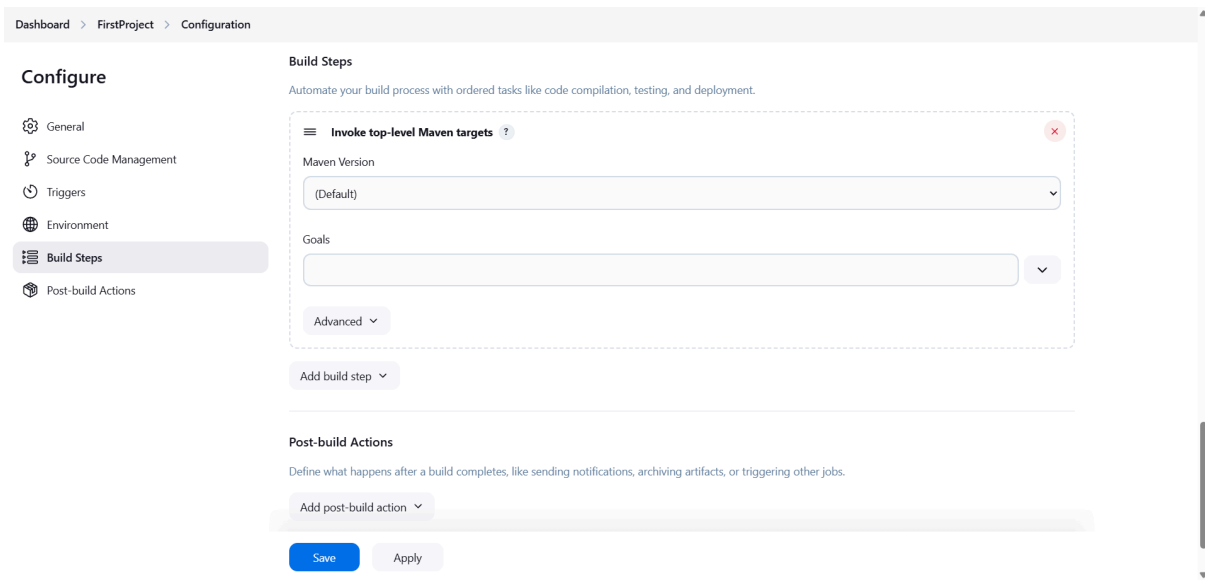
Step 5: Scroll Down and go to BRANCHES TO BUILD , write the branch name there in which you want to work

The screenshot shows the 'Branches to Build' configuration page. The sidebar is the same as in the previous screenshot. The main area is titled 'Branches to build' and includes an 'Add Repository' button. Below this, there is a 'Branch Specifier (blank for 'any')' field with the value '*/main'. There is an 'Add Branch' button. Below that, there is a 'Repository browser' dropdown menu showing '(Auto)'. At the bottom, there is an 'Additional Behaviours' section with an 'Add' button and a 'Triggers' section with a description: 'Set on automated actions that start your build based on specific events like code changes or scheduled times'. 'Save' and 'Apply' buttons are at the bottom.

Step 6: Click ADD BUILD STEPS -----> SELECT INVOLVE-LEVEL MAVEN TARGETS



Step 7: In MAVEN VERSION select maven



Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

≡

Invoke top-level Maven targets ?

✕

Maven Version

(Default)

(Default)

maven

Advanced ▾

Step 8: In GOALS type **test -DsuiteFile=packages**

Dashboard > FirstProject > Configuration

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

≡

Invoke top-level Maven targets ?

✕

Maven Version

maven

Goals

test -DsuiteFile=packages

Advanced ▾

Add build step ▾

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Add post-build action ▾

Save Apply

Click APPLY and then Click Save

Now go to the Dashboard and you will see Your Created job there

Dashboard >

+ New Item

Build History

Manage Jenkins

My Views

Build Queue

No builds in the queue.

Build Executor Status

0/2

All +

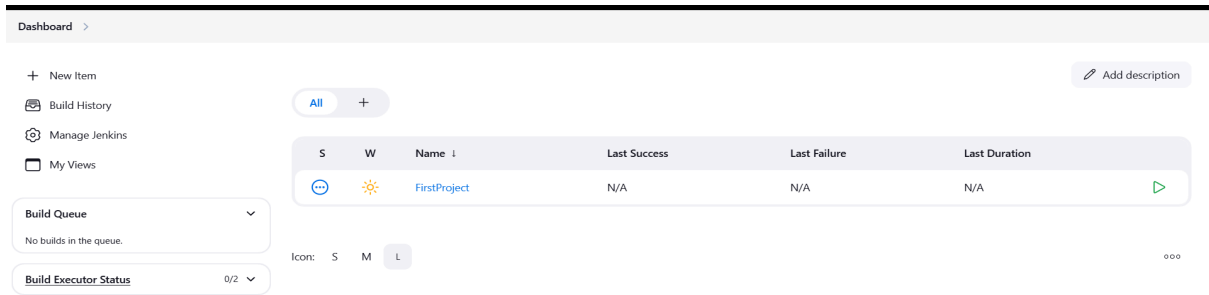
Add description

S	W	Name	Last Success	Last Failure	Last Duration
☑	☀	FirstProject	N/A	N/A	N/A

Icons: S M L

HOW TO RUN A JOB

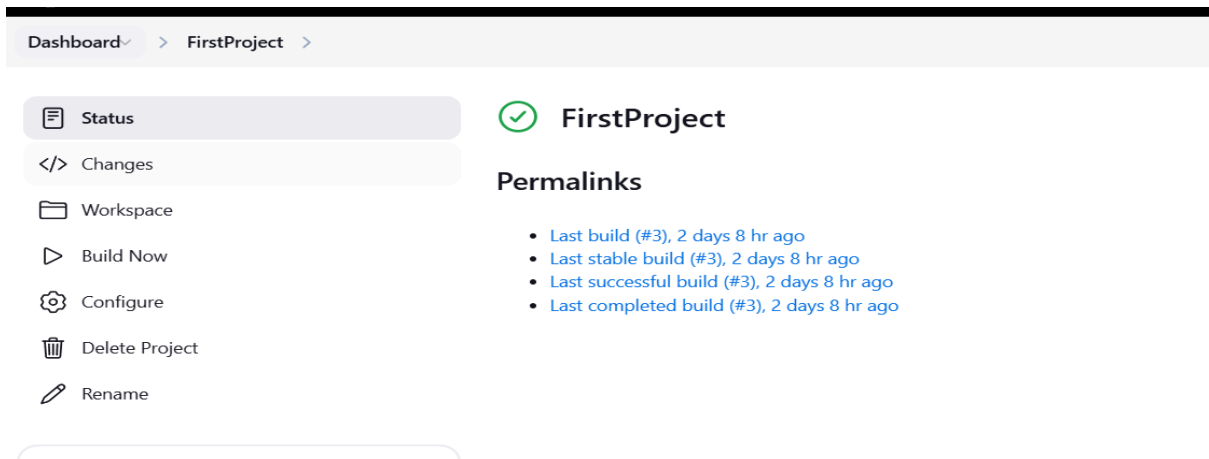
Step 1: Open Your Job that you want to run



The screenshot shows the Jenkins Dashboard. On the left, there's a sidebar with links: '+ New Item', 'Build History', 'Manage Jenkins', and 'My Views'. Below these are two expandable sections: 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing '0/2'). The main area features a table with columns: 'S', 'W', 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. A single row is visible for 'FirstProject' with status icons (a blue circle with a white 'C' and a yellow sun) and 'N/A' values for the last success, failure, and duration. A green play button icon is on the right of the row. Above the table is a filter bar with 'All' and a plus sign. Below the table is a legend for status icons: 'S' (Success), 'M' (Missing), and 'L' (Failed). A 'Add description' link is in the top right corner.

S	W	Name	Last Success	Last Failure	Last Duration
		FirstProject	N/A	N/A	N/A

Step 2; Click on build now present on left side



The screenshot shows the Jenkins Job View for 'FirstProject'. The left sidebar contains a list of actions: 'Status' (selected), 'Changes', 'Workspace', 'Build Now' (highlighted with a blue bar), 'Configure', 'Delete Project', and 'Rename'. The main area shows a green checkmark icon next to the job name 'FirstProject'. Below this is a 'Permalinks' section with a list of links: 'Last build (#3), 2 days 8 hr ago', 'Last stable build (#3), 2 days 8 hr ago', 'Last successful build (#3), 2 days 8 hr ago', and 'Last completed build (#3), 2 days 8 hr ago'. The breadcrumb at the top reads 'Dashboard > FirstProject'.

Step 3: Click on Build at the bottom and then in console output you can see the output



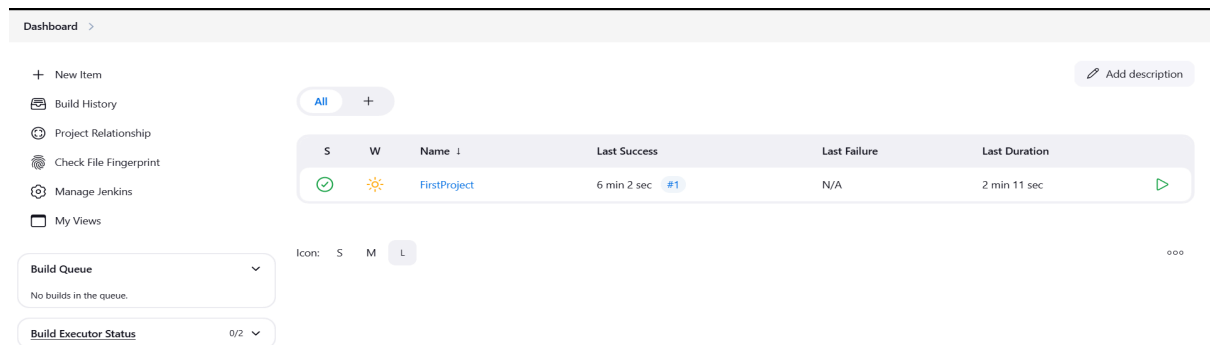
The screenshot shows the Jenkins Console Output for the 'FirstProject' job. The breadcrumb at the top reads 'Dashboard > FirstProject > #1 > Console Output'. The output text is as follows:

```
at 36 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-grouper/3.0.0-M7/surefire-grouper-3.0.0-M7.jar (40 kB at 52 kB/s)
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running TestSuite
Sample Testing
Void b
Void c
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.78 s - in TestSuite
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 01:42 min
[INFO] Finished at: 2025-03-16T15:45:20+05:30
[INFO]
Finished: SUCCESS
```

At the bottom right, there are links for 'REST API' and 'Jenkins 2.492.2'.

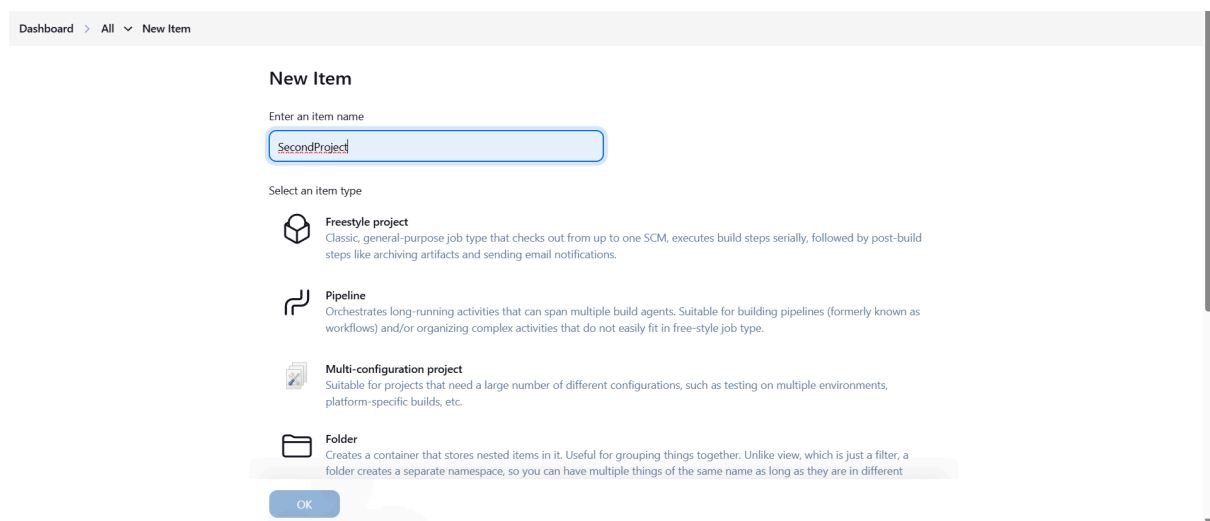
CREATING ANOTHER JOB BY COPYING THE JOBS THAT ARE ALREADY PRESENT

Step 1: Go to DashBoard -----> click NEW ITEM



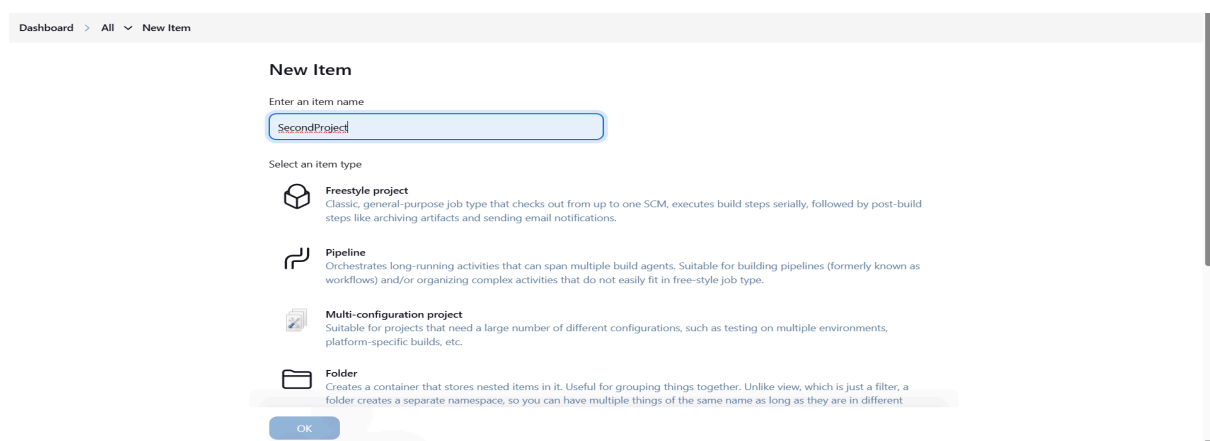
The screenshot shows the Jenkins Dashboard. On the left is a sidebar with navigation links: '+ New Item', 'Build History', 'Project Relationship', 'Check File Fingerprint', 'Manage Jenkins', and 'My Views'. Below these are two summary boxes: 'Build Queue' (No builds in the queue) and 'Build Executor Status' (0/2). The main area displays a table of jobs. The first job is 'FirstProject', which is successful (green checkmark), has a sun icon, and a last success time of '6 min 2 sec'. A table with 7 columns is shown: S, W, Name, Last Success, Last Failure, Last Duration, and a green play button icon. The row for 'FirstProject' contains: a green checkmark, a sun icon, 'FirstProject', '6 min 2 sec', '#1', 'N/A', and '2 min 11 sec'. Above the table is a filter bar with 'All' and a plus icon. To the right of the table is a link 'Add description'. Below the table is a section for 'Icon' with options 'S', 'M', and 'L'.

Step 2: Write the name of the item (2nd job)



The screenshot shows the 'New Item' form in Jenkins. The breadcrumb is 'Dashboard > All > New Item'. The form has a title 'New Item'. Below it is a text input field 'Enter an item name' with the value 'SecondProject'. Below that is a section 'Select an item type' with four options: 'Freestyle project' (Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.), 'Pipeline' (Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.), 'Multi-configuration project' (Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.), and 'Folder' (Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different namespaces). At the bottom is an 'OK' button.

Step 3 : Scroll Down & write the name of the JOB which you want to copy & click ok



This screenshot is identical to the previous one, showing the 'New Item' form in Jenkins. The breadcrumb is 'Dashboard > All > New Item'. The form has a title 'New Item'. Below it is a text input field 'Enter an item name' with the value 'SecondProject'. Below that is a section 'Select an item type' with four options: 'Freestyle project' (Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.), 'Pipeline' (Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.), 'Multi-configuration project' (Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.), and 'Folder' (Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different namespaces). At the bottom is an 'OK' button.

5 Creates a set of Pipeline projects according to detected branches in one SCM repository.

 **Organization Folder**
Creates a set of multibranch project subfolders by scanning for repositories.

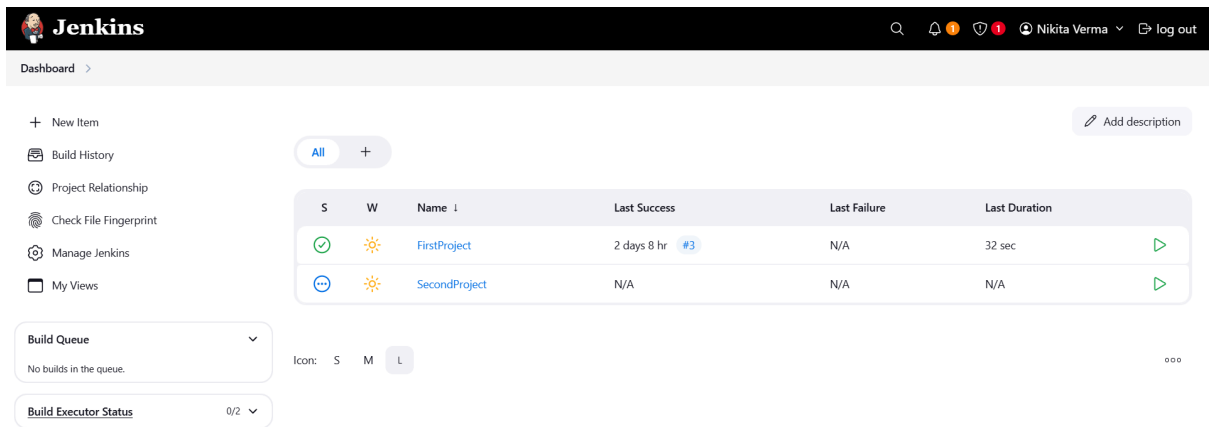
If you want to create a new item from other existing, you can use this option:

Copy from

FirstProject

OK

Now you can see your created job in the Dashboard

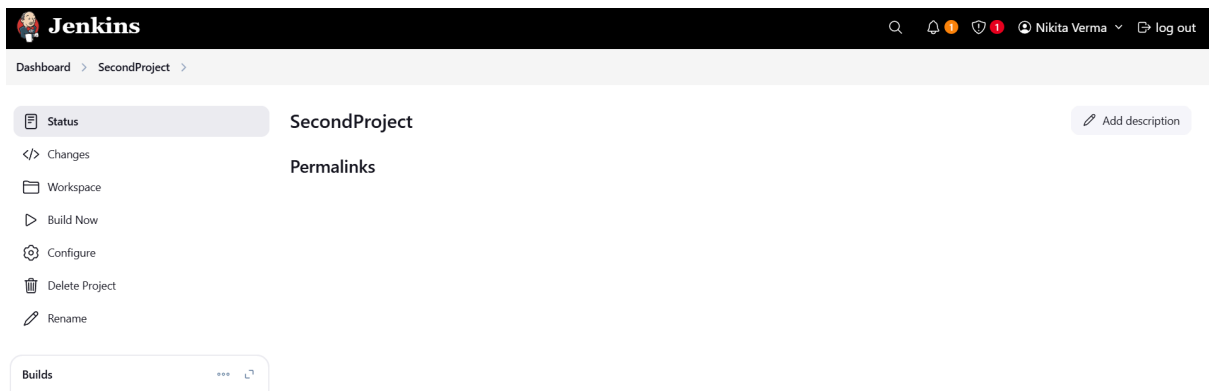


The screenshot shows the Jenkins Dashboard. The top navigation bar includes the Jenkins logo, a search icon, notification icons, and the user profile 'Nikita Verma' with a 'log out' link. The main content area has a left sidebar with links: '+ New Item', 'Build History', 'Project Relationship', 'Check File Fingerprint', 'Manage Jenkins', and 'My Views'. The main panel displays a table of jobs. The table has columns: 'S' (Status), 'W' (Web icon), 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. Two jobs are listed: 'FirstProject' and 'SecondProject'. 'FirstProject' shows a success status with a green checkmark and a duration of '32 sec'. 'SecondProject' shows a pending status with a blue circle and 'N/A' for duration. Below the table, there are filters for 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing '0/2').

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	FirstProject	2 days 8 hr #3	N/A	32 sec
⋯	☀	SecondProject	N/A	N/A	N/A

DELETING A JOB

Step 1: Open you Job by clicking it, on left side you can see Delete option, delete From There



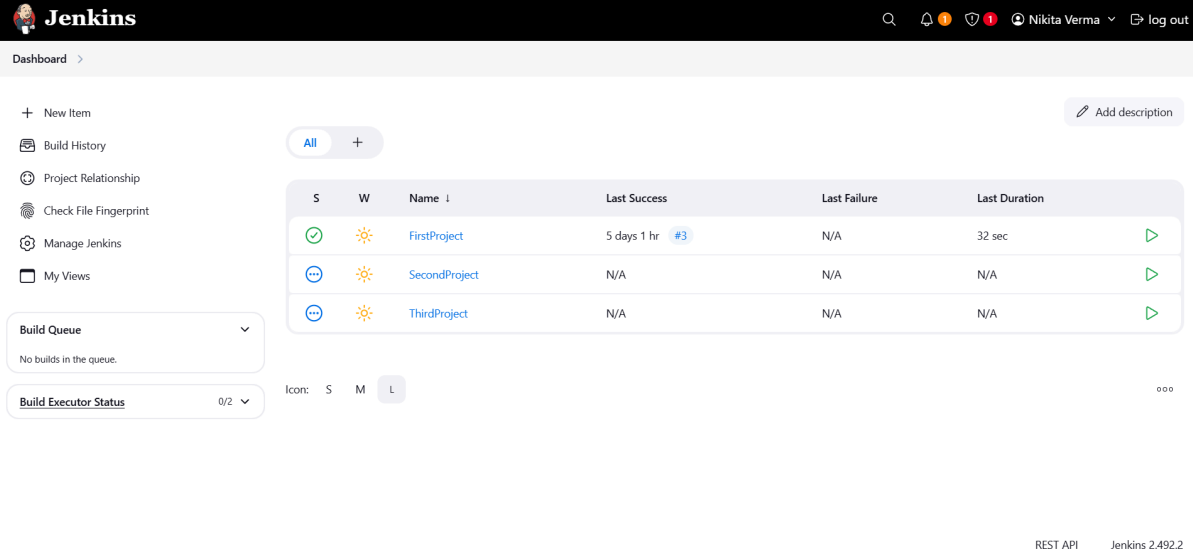
The screenshot shows the Jenkins Job View for 'SecondProject'. The top navigation bar is the same as the dashboard. The breadcrumb trail is 'Dashboard > SecondProject'. The left sidebar contains links: 'Status' (selected), '</> Changes', 'Workspace', 'Build Now', 'Configure', 'Delete Project', and 'Rename'. The main panel shows the job name 'SecondProject' and a 'Permalinks' section. At the bottom, there is a 'Builds' section with a list of builds and a 'Build Now' button.

BASIC JENKINS PIPELINE

A basic Jenkins Pipeline is like : if one job is completed then the next job starts Automatically

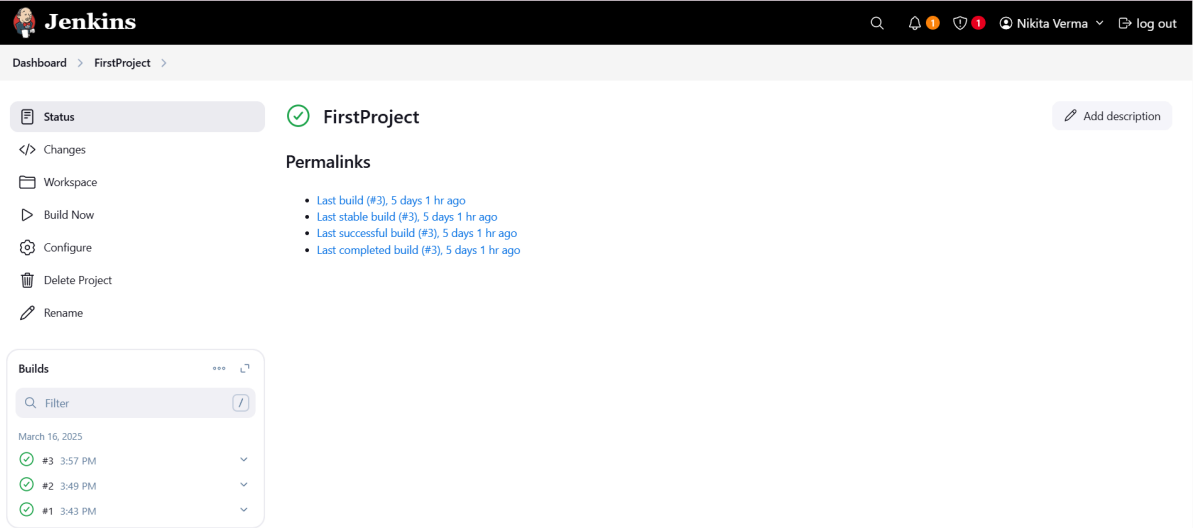
For a basic jenkins pipeline to work there should be minimum 2 jobs

Step1 : In Dashboard, go to your 1st Project -----> Click Configure



The screenshot shows the Jenkins Dashboard. The top navigation bar includes the Jenkins logo, a search icon, notification icons, and the user name 'Nikita Verma' with a 'log out' link. The main content area features a sidebar on the left with links for 'New Item', 'Build History', 'Project Relationship', 'Check File Fingerprint', 'Manage Jenkins', and 'My Views'. Below these are two summary boxes: 'Build Queue' (No builds in the queue) and 'Build Executor Status' (0/2). The main area displays a table of jobs. The first job, 'FirstProject', is in a successful state (green checkmark) and has a last success time of '5 days 1 hr #3' and a duration of '32 sec'. The other two jobs, 'SecondProject' and 'ThirdProject', are in a pending state (blue circles with dots). At the bottom right, it says 'REST API' and 'Jenkins 2.492.2'.

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☀	FirstProject	5 days 1 hr #3	N/A	32 sec
...	☀	SecondProject	N/A	N/A	N/A
...	☀	ThirdProject	N/A	N/A	N/A



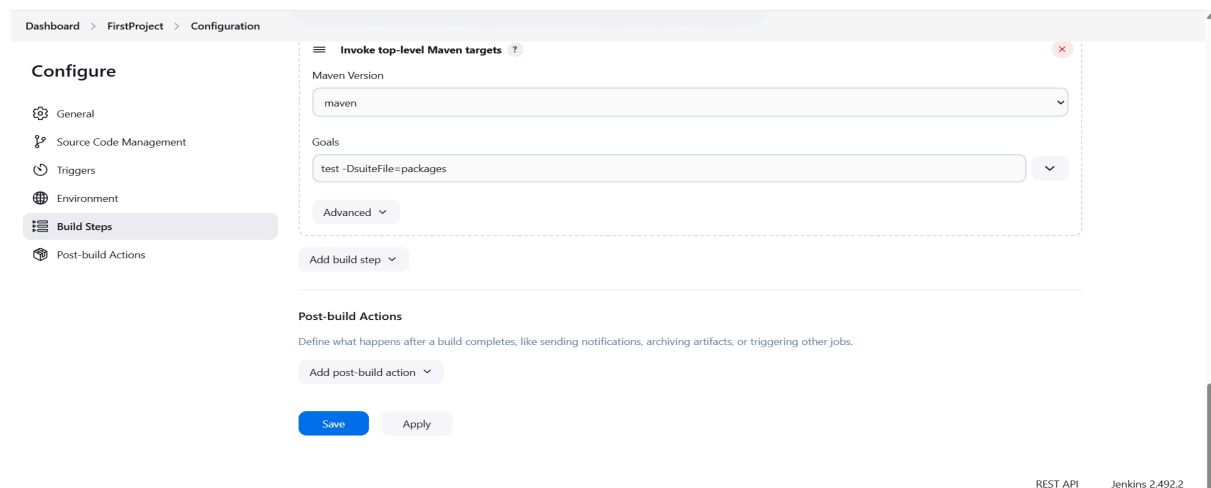
The screenshot shows the Jenkins configuration page for a project named 'FirstProject'. The top navigation bar is the same as the dashboard. The main content area has a sidebar on the left with links for 'Status', 'Changes', 'Workspace', 'Build Now', 'Configure', 'Delete Project', and 'Rename'. The 'Status' link is selected. The main area shows the project name 'FirstProject' with a green checkmark icon. Below this is a 'Permalinks' section with a list of links for the last build, last stable build, last successful build, and last completed build, all of which are '#3' and occurred '5 days 1 hr ago'. At the bottom left, there is a 'Builds' section with a filter input and a list of builds for 'March 16, 2025'. The builds are listed as '#3 3:57 PM', '#2 3:49 PM', and '#1 3:43 PM', each with a green checkmark icon.

Builds

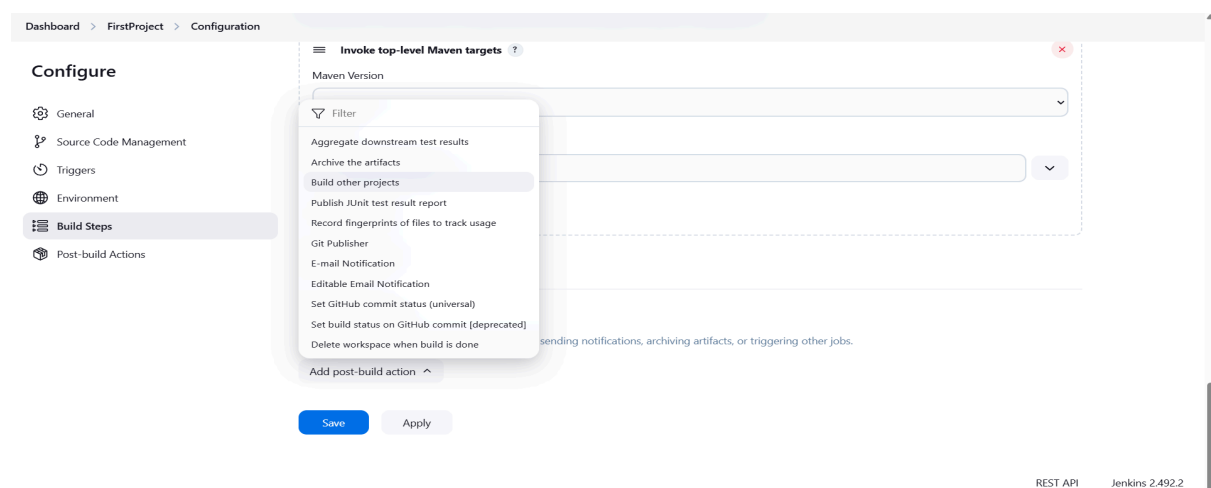
March 16, 2025

- ✓ #3 3:57 PM
- ✓ #2 3:49 PM
- ✓ #1 3:43 PM

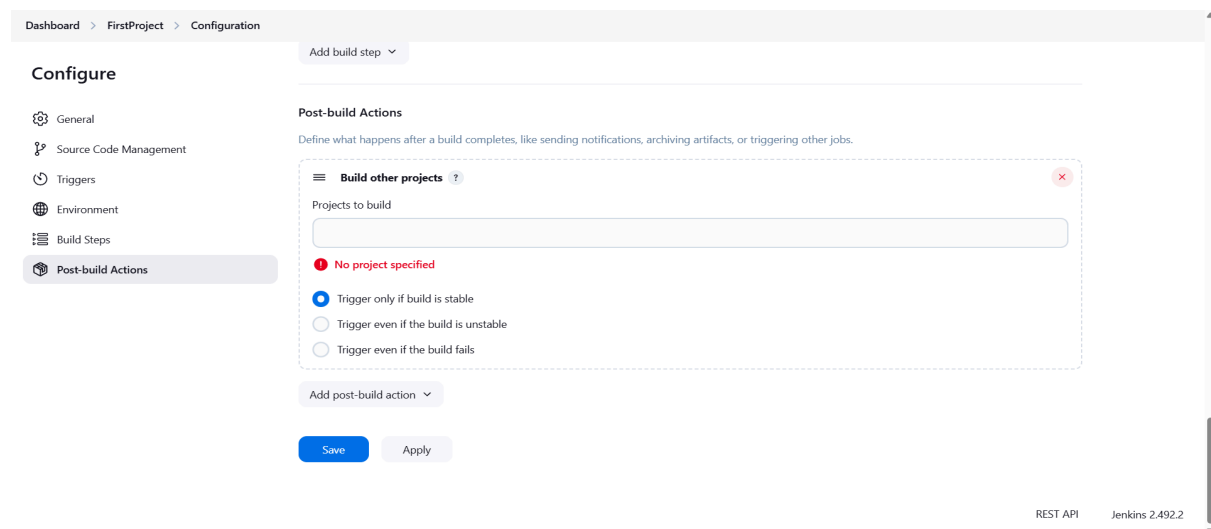
Step 2 : Scroll Down and go to Post Build Actions and Click On Add Build Actions



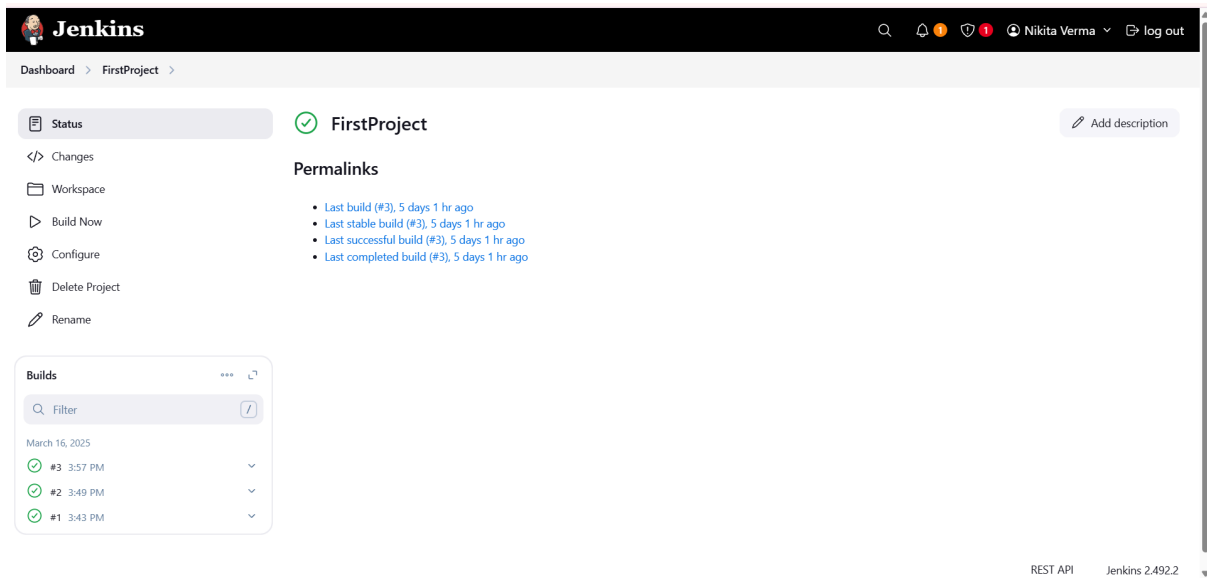
Step 3 : Select BUILD other Projects



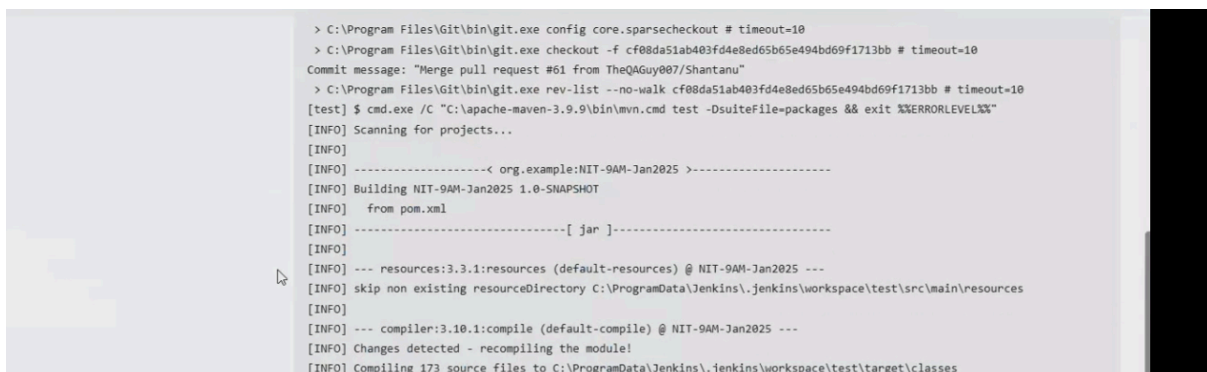
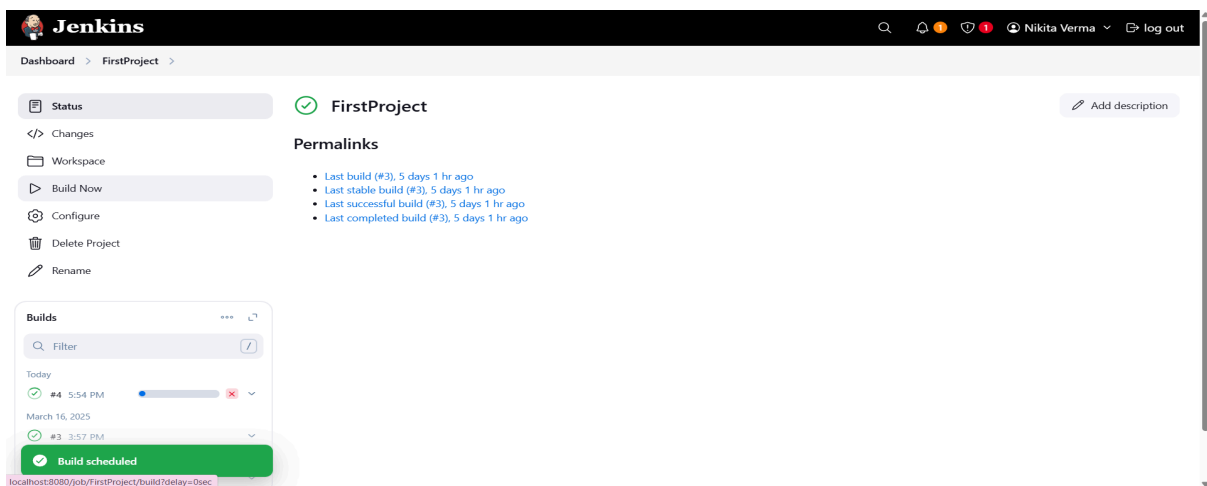
Step 4 : Write the name of your Job in “ project to Build” section that you want to get executed next. Then Click SAVE



Step 5 : Now go to 1st Job and Click Build Now

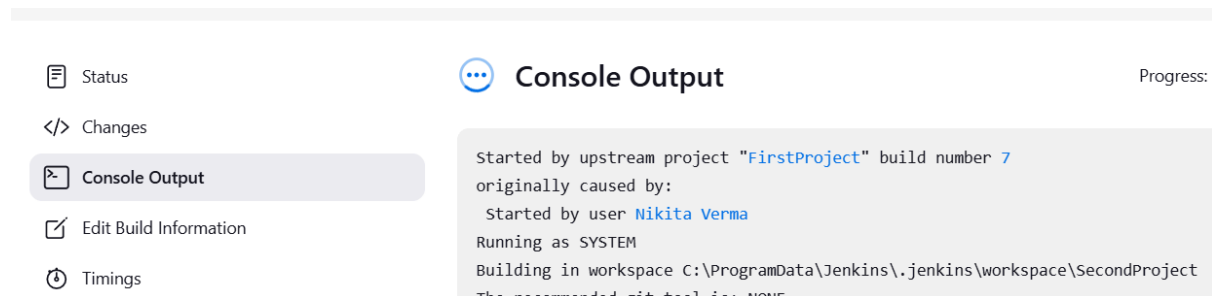


Step 6 : Your Build will start, click on that build number and in console you will see it



Once this job completes its execution immediately the next Build that you have added will start executing automatically.

In the console you can also see the message on the top “ Started by upstream Project FirstProject ...”



The screenshot shows the Jenkins interface. On the left, there is a sidebar with navigation links: Status, Changes, Console Output (highlighted), Edit Build Information, and Timings. The main area is titled 'Console Output' and displays the following text: 'Started by upstream project "FirstProject" build number 7 originally caused by: Started by user Nikita Verma Running as SYSTEM Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\SecondProject The recommended git tool is: NONE'.

This is how a basic pipeline works.

Jenkins: Cron Syntax:

Used for Scheduling Jobs in Jenkins.

The order of the Syntax is as follows:

Minute Hour Day(Month) Month Day(Week)

To Schedule using Cron Syntax do the following:

- Open the Job -> Configure -> Build Triggers -> Build Periodically -> Enter the appropriate Cron Syntax

Example Cron Syntax below:

0 10 * * * - Every day at 10 AM.

0 10 30 * * - Every 30th of the month at 10 AM.

0 0 * * 0 - 12 AM Every Sunday of the week.

